

Settings for TRexFitter blocks

Standard fit

The following settings are for normal fits, performed without the action `m`. The `Job` block subcategories are provided only to make the reading of the settings easier and *should not* be used in the config file.

`Job` block settings

Option	Function
General	
ExperimentLabel	the name of the experiment (e.g. "ATLAS"). The default is <code>ATLAS</code>
Label	the label which will be shown on plots, if 'NONE' is set, no label will be shown
POI	the name of the parameter of interest; this should correspond to a NormFactor defined below. It is possible to define more than one POI, as a comma-separated list
ReadFrom	can be HIST or NTUP; default is HIST
Lumi	value to scale all the "NormalizedByTheory" samples
LumiScale	additional value to scale 'after' histogram creation (for fast scaling) IMPORTANT: use it only if you know what you are doing!!
DebugLevel	0 = prints only Warning and Errors, 1 = additionally prints Info messages, 2 = additionally prints Debug

Option	Function
	messages, >2 additionally prints Verbose messages. For option <2 RooFit/Roostats messages will be heavily suppressed
LoggingFormat	Can be set to 'NONE' to have no logging header. Otherwise a list of options from: 'TIME','DATE', 'FILELINES' and 'FUNCTION' is allowed. Each option turns on the logging header for each line presenting, in order, the time of the day, the day, name of the file and lines where the message is printed from, the function name where the message is printed from. Default is 'TIME,DATE,FILELINES'. Please, make sure there are no white spaces between the commas as the info is pre- read.
Logo	is set to TRUE will print the <code>TRExFitter</code> logo
HistoChecks	NOCRASH: means that if an error is found in the input histograms, the code continues (with only warnings) -- default leads to a crash in case of problem, if set to NOCRASH, also prints warning instead of error (and crash) when input files are not found for the histogram building step
SplitHistoFiles	set this to TRUE to have histogram files split by region (useful with many regions and/or run in parallel)
ImageFormat	png, pdf or eps
SmoothingOption	Choose which smoothing option to use, allowed parameters are: MAXVARIATION (default), TTBARRESONANCE (see also FAQ section), COMMONTOOLSMOOTHMONOTONIC, COMMONTOOLSMOOTHPARABOLIC, TCHANNEL, KERNELRATIOUNIFORM, KERNELDELTAGAUSS or KERNELRATIOGAUSS.

Option	Function
CorrectNormForNegativeIntegral	By default, if there are samples with negative yields in some bins, the total integral over that sample will be re-scaled after the yield in the negative bins was corrected to 1e-6, such that the total yield across the sample in this region is consistent with the yield before fixing the negative bins. If this option is set to TRUE (by default, it is FALSE), then this re-scaling will also happen if the total yield initially was negative. This can lead to unexpected behavior (expert option, proceed with caution).
MergeUnderOverflow	if set to TRUE, the underflow content of each histogram is added to the first bin and the overflow to the last one (default is FALSE for HIST inputs and TRUE for NTUP inputs)
SuppressNegativeBinWarnings	If set to TRUE, will suppress warning messages about negative or 0 content in bins
Bootstrap	(only works with NTUP inputs) if set, the bootstrap method will be used; the argument should be a string like <pre>bsWeight(BootstrapIdx,eventNumber,mcChannelNumber)</pre> , where <code>bsWeight</code> should be loaded with <code>CustomFunctions: "bsWeight.C"</code> and <code>eventNumber</code> and <code>mcChannelNumber</code> should be existing branches for all the MC ntuples; then, to produce the <i>i</i>-th bootstrap pseudo-experiment, or to run on it (e.g. to perform a fit) the command-line option <code>BootstrapIdx=<i></code> should be given, with <code><i>=0,1,2,3...</code>. Alternatively there is default function <code>util/BootstrapDefault.C</code> which would be called as <code>BootstrapDefault(eventNumber+mcChannelNumber+BootstrapIdx)</code>.
BootstrapSyst	Specifies the systematics for which should be the bootstrap done.

Option	Function
BootstrapSample	Specifies the sample for which should be the bootstrap done. Nominal and all correlated systematics are smeared, correlated systematics are specified using IsCorrelated option.
BootstrapNomHistos	Specifies a directory in which histograms of systematics/samples that aren't bootstrapped live. The code will look for non bootstrapped histograms in this directory if provided, otherwise it will look for them in the same directory as the config's histograms. This is useful to run the <code>n</code> step only for the bootstrapped systematic/sample.
ReplacementFile	allows usage of placeholders in the config, which will be overwritten by values provided in an external file; see Replacement file section in the README
CustomIncludePaths	can be used in case the user wants to add include path(s) needed when reading custom-functions (same as typing <code>.include /path/to/include</code> before calling <code>.L myMacro.C</code>)
AllowWrongRegionSample	Can be TRUE or FALSE (default). When set to TRUE code will print only warnings when chosen samples or regions for various options are not defined. When set to FALSE the code will print errors and stop when the samples/regions are not defined.
MaxNtupleEvents	valid only for option NTUP; if set to N, only first N entries per ntuple read (useful for debugging)

Option	Function
CustomFunctions	list of .C files with definition and implementation of functions to be used in strings defining selections or weights (see this link: https://wiki.physik.uzh.ch/lhcb/root:ttreedraw , notice that the file and function names should match and that all the arguments of the function should have default values)
CustomFunctionsExecutes	semicolon separated list of functions to be executed right after the loading .C files, in case of any initialization step required before filling ntuples (can be set via a command line option 'CustomFunctionsExecutes')
MCweight	only for option NTUP; string defining the weight (for MC samples only)
Selection	only for option NTUP; string defining the selection
DoExtendedCovariances	If set to <code>TRUE</code> (default is <code>FALSE</code>) will produce covariance matrices for each systematic and the stat only part as well as the linear sum of these matrices. To be able to do this, you need to run the usual fit, then run with <code>StatOnlyFit=TRUE</code> in the command line and rerun the <code>f</code> step (this is needed for the stat only covariance matrix) and then you need to run the usual ranking (this is needed for the systematic covariance matrices). The matrices are created during the plotting part of the ranking and are stored in <code>Covariances/</code> folder. When using the <code>HEPDataFormat: TRUE</code> setting, will also produce the HEPData format for these matrices.
Paths	

Option	Function
HistoPath(s)	valid only for option HIST above is selected; it's the path(s) where the input root files containing the histograms are stored
HistoFile(s)	valid only for option HIST; it's the file name(s) where the input root files containing the histograms are stored
HistoName(s)	valid only for option HIST; it's the histogram name(s) to read from the file(s)
HistoNameNominal	valid only for option HIST; name of the nominal histogram, in case the systematic histogram names cannot be build by suffixing but instead replace the nominal name. Behaves like HistoNameSuff for Systematics if Nominal was a Systematic
NtuplePath(s)	valid only for option NTUP; it's the path(s) where the input root files containing the ntuples are stored
NtupleFile(s)	valid only for option NTUP; it's the file names(s) where the input root files containing the ntuples are stored
NtupleName(s)	valid only for option HIST; it's the tree name(s) to read from the file(s)
FriendPath(s)	valid only for option UseFriend is true; it's the path(s) where the input root files containing the ntuples to be added as friend to NtuplePaths
InputFolder	specify it to read fit input histograms from a different directory than <code><jobName>/Histograms/</code>
InputName	specify it to read fit input histograms from files with different name than <code><jobName>_blabla.root</code>
OutputDir	specify it to write everything in a different directory, the full directory structure will be: "OutputDirName/" (if set)

Option	Function
	+ jobname (can be set via a command line option 'Job')
WorkspaceFileName	if specified, an external ws can be used as input for fitting (not 100% supported)
Suffix	added to file names of plots, workspace, fit results etc. (equivalent to command line option)
SaveSuffix	added to file name of histograms, for usage with hupdate (equivalent to command line option)
SummaryPrefix	adds a prefix to summary and merge plots
ResponseMatrixName(s)	Name(s) of the histogram for response matrix
ResponseMatrixFile(s)	File path(s) of the histogram for response matrix
ResponseMatrixPath(s)	Folder path(s) of the histogram for response matrix
ResponseMatrixNameNominal	Nominal histogram name
AcceptanceName(s)	Name(s) of the histogram for acceptance
AcceptanceFile(s)	File path(s) of the histogram for acceptance
AcceptancePath(s)	Folder path(s) of the histogram for acceptance
AcceptanceNameNominal	Nominal histogram name
SelectionEffName(s)	Name(s) of the histogram for selection efficiency
SelectionEffFile(s)	File path(s) of the histogram for selection efficiency
SelectionEffPath(s)	Folder path(s) of the histogram for selection efficiency
SelectionEffNameNominal	Nominal histogram name

Option	Function
MigrationName(s)	Name(s) of the histogram for migration matrix
MigrationFile(s)	File path(s) of the histogram for migration matrix
MigrationPath(s)	Folder path(s) of the histogram for migration matrix
MigrationNameNominal	Nominal histogram name
UseFolderStructure	If set to <code>TRUE</code> (default) will store the histograms produced by TRExFitter using "RegionName/SampleName/SystematicName" folder structure. This should make the retrieving of the histograms much faster in case of many histograms. If set to <code>FALSE</code> will use the old approach with no folders - needed for backward compatibility
RecreateBinningFiles	If set to <code>TRUE</code> (default is <code>FALSE</code>) will tell the code to overwrite the txt files in the 'Binnings' folder that store the number of bins for each region. These numbers are needed for some of the steps when the histograms are not read (to speed up the execution). You should set this to <code>TRUE</code> whenever you change the number of bins in your config file/input. Can be called from the command line as 'RecreateBinningFiles=TRUE'
WorkspaceCreationType	This option allows to control which workspaces are created during the <code>w</code> step. In general, two workspaces are needed: one for the fit (includes SRs and CRs) and one for plotting with all regions (including the VRs). The code will not run the plotting WS if there are no validation regions. The option can be used to produce only the WS for the fit (set to <code>FORFIT</code>), only for the plotting (set to <code>FORPLOTS</code>) or both WSs (set to <code>BOTH</code> - default). This option can be set via the command line.
SeparationPlot	list of comma separated samples to add in the separation plot (option "a").

Option	Function
HEPDataFormat	If set to TRUE (default is FALSE), will produce outputs in HEPData format
UseHEPDataRounding	If set to TRUE (default is FALSE), will apply rounding the HEPData values and errors
UheppFormat	If set to TRUE (default is FALSE), will produce plots in Uhepp format
CombinerFormat	If set to TRUE (default is FALSE), will produce outputs in Combiner yaml format when running <code>Ranking=plot</code>
ProducePerRegionWS	If set to TRUE (default is FALSE), will produce per-region WSs. These are not needed in TRExFitter anymore, but it might be needed for backwards-compatibility
Pruning	
PruningType	Can be set to <code>BACKGROUNDREFERENCE</code> , <code>COMBINEDREFERENCE</code> or <code>COMBINEDSIGNAL</code> (default is <code>SEPARATESAMPLE</code>), and pruning (both shape and norm) will be done w.r.t. background/total./signal only. The <code>COMBINEDSIGNAL</code> can be used for UNFOLDING fit. Works also when <code>SeparateGammas</code> is used. When <code>COMBINEDSIGNAL</code> is used, it is considered for <code>SeparateGammas</code> , but for other NPs it reverts to <code>SEPARATESAMPLE</code> .
SystPruningShape	Lower threshold to remove a shape systematic from the fit/limit (suppression is done per sample and per region) (e.g.: 0.02 for 2%). The "shape" pruning is performed by taking the alternative distribution (the up/down template) and normalizing it in each region to the same yield as the nominal distribution (just for the purpose of performing pruning). If the alternative template then differs by more than the threshold from the nominal template in any bin, the "shape" component

Option	Function
	will not be pruned. A non-prunable "shape" therefore is anything where at least one bin varies more than the threshold from the average.
SystPruningNorm	Lower threshold to remove a normalisation systematic from the fit/limit (suppression is done per sample and per region) (e.g.: 0.02 for 2%)
SystLarge	All systematics above this threshold will be removed, unless <code>RemoveLargeSyst</code> is set to <code>FALSE</code> , then the systematics will be flagged in the pruning plot) (e.g. 0.4 will flag systematics that are larger than 40%). The default is -1, meaning no pruning.
RemoveLargeSyst	Will prune systematics above threshold set by <code>SystLarge</code> if set to <code>TRUE</code> (default is <code>FALSE</code>). It will remove both shape and norm component, effectively removing the systematic uncertainty for a given sample.
RemoveSystOnEmptySample	If set to <code>TRUE</code> , will prune systematics for samples where nominal has integral $< 1e-4$. Default is <code>FALSE</code>
PruningShapeOption	Can be <code>MAXBIN</code> (Default) or <code>KSTEST</code> . <code>MAXBIN</code> checks if the systematic variation has at least one bin which is different compared to the nominal (variation is $\geq$ threshold). <code>KSTEST</code> runs the Kolmogorov test (using the <code>X</code> option, a.k.a running pseudoexperiments). When using <code>KSTEST</code> , if the probability is $\leq (1-\text{threshold})$ then the systematic is marked as having shape component.
ShowValidationPruning	If set to <code>TRUE</code> will show the pruning also in validation regions. Default is <code>FALSE</code> .
KeepPruning	if set to <code>TRUE</code> , the first time the ws is created (option w) a Pruning.root file is created under <code><jobName>/</code> and used for future operations to skip pruned systematics

Option	Function
	(makes operations much faster in case many syst are pruned)
DrawPruningPlot	if set to TRUE (default), the code will produce the pruning plot. Setting it to FALSE can save a lot of CPU and RAM for setups with many regions and samples
Affecting Fit	
IntCode	interpolation code (should match the one used in RooStats). Only code 4 (default) and 0 is currently supported for plotting.
MCstatThreshold	by default, the MC stat uncertainty is included in the fit (and to the plots); a NP will be added for each bin with an MC stat uncertainty > this threshold (relative) if the option is set to a float (default: no threshold); can also set to NONE in order to disable MC stat uncertainty completely. If set to <code>NONE</code> , it will also disable the MC stat uncertainty from all samples with <code>SeparateGammas: TRUE</code> , effectively ignoring this option
MCstatConstraint	constraint used for MC stat uncertainties, can be set to 'GAUSSIAN' or 'POISSON' (default)
StatOnly	the code ignores systematics and MC stat uncertainties from all computations (limits, significances, fit, ...); need to re-create ws in case of limit and significance
FixNPforStatOnly	if set to TRUE, when running stat-only (with either of the two options) also the norm factors other than the POI are kept fixed
CustomAsimov	if set, the workspace will be created with an AsimovData built according to Sample- > <code>AsimovReplacementFor</code> option (see below) instead of

Option	Function
	data. Possible to override in commandline via CustomAsimovOverride=dataset
WriteCustomAsimovToWS	if set to TRUE, write out workspace containing custom Asimov data defined by CustomAsimov
GuessMCStatEmptyBins	For bins with negative yields, the yield is corrected to 1e-06. If a stat. uncertainty on that bin was defined, it is kept. If the stat. uncertainty was however 0, then this option kicks in. If it is set to TRUE (default), the smallest stat. uncertainty from any other bin in the distribution will be used. If set to FALSE, or if all other bins have no stat. uncertainty defined, a 100% uncertainty (of 1e-06) is applied instead.
DecorrSysts	comma-separated list of systematics which you want to decorrelate from another channel (this is done by automatically attaching a suffix to the NuisanceParameter for each of them); can use wildcards
DecorrSuff	the suffix to attach when using DecorrSysts
SmoothMorphingTemplates	if set to TRUE (default is FALSE), the templates used for morphing are forced to have linear dependence on the morphing parameter, bin-by-bin (plots are produced per bin, in the Morphing directory)
PropagateSystsForMorphing	if set to TRUE (default is FALSE), the non-nominal templates inherit all the systematics from the nominal template; NB: the nominal template is determined by the nominal value of the norm factor in the config
ExcludeFromMorphing	The specified sample is left out from the morphing (useful to do closure tests for morphing).
AlternativeShapeHistFactory	For systematic uncertainties defined via a single template and then symmetrized, there are two ways of

Option	Function
	building the (normalized) <code>HistoSys</code> template that enters <code>HistFactory</code> . By default, <code>TRExFitter</code> first symmetrizes the provided template, and then normalizes both resulting templates. If this option is set to TRUE (false by default), then the order of operations is switched. <code>TRExFitter</code> will first normalize the provided template to the same yield as nominal, and then symmetrize the resulting template. This means that the templates passed to <code>HistFactory</code> are symmetric, while by default they are not completely symmetric. (Note that this does not mean that the normalization effect is dropped from the systematic, it is handled separately in <code>HistFactory</code> as an <code>OverallSys</code> .
ApplyGammaCorrection	If set to TRUE (default is TRUE) will apply a correction to the gammas used in "CorrelateGammasWithSample" to remove the double counting when samples come from different sources. Set to FALSE if you generate the samples e.g. by reweighting.
ToysForErrorBand	If set to a value N that is > 0 (default is -1) will run N toys to estimate the post-fit error band. In this case, the error band will be, generally speaking, asymmetric.
Cosmetics	
UseGammaPulls	if set to TRUE (Default), the fit results in terms of gamma parameter pulls, constraints and correlations are propagated to the post-fit plots, when possible (i.e. not for validation plots of course)
PlotOptions	a set of options for plotting: YIELDS: if set, the legend will be one-column and will include the yields; otherwise two-columns and no yields NORMSIG: add normalised signal to plots NOSIG: don't show signal in stack SKIPSIG: draw uncertainty band for backgrounds, then add signal to stack

Option	Function
	OVERSIG: overlay signal (not normalised) CHI2: the chi2/ndf and chi2 prob will be printed on each plot, provided that the option GetChi2 is set PREFITONPOSTFIT: draw a dashed line on the postfit plot that indicates the sum of prefit background XERR: adds the horizontal error bars on the data and the ratio plots LEFT: forces legend entry to be left-aligned when adding yields to legend RIGHT: forces legend entry to be right-aligned even when not adding yields to legend OPRATIO: use kOpenCircle as marker style in ratio instead of the default kFullCircle NORATIO: do not draw ratio pad KEEPDATAERRORS: Use the BinErrors from the histogram from the data uncertainty per bin
OVERSIGSCALE	If the option "OVERSIG" has been set in "PlotOptions", the overlaid signal histogram is scaled by a constant factor. The scale of the signal histogram remains the same pre- and post-fit, and the post-fit value of the POI is not taken into account in the plotted overlaid signal. However, note that this scale is not applied in the Summary and Merge plots.
POIUnit	a unit can be added to the POI, for cosmetic reasons, in case it's not a pure number. In case of more than one POI, the argument should be in the form <code>"name-of-poi-1": "unit-1", "name-of-poi2": "unit-2"</code>
PlotOptionsSummary	the same as PlotOptions but for the summary plot (if nothing is specified, PlotOptions is used)
RatioYtitle	Label to be used on the Y-axis of the ratio plot, default is "Data/pred."
TableOptions	a set of options for tables: STANDALONE: default! If not set, no

Option	Function
	<pre>"\begin{document}" FOOTNOTESIZE: -> \footnotesize LANDSCAPE: -> \begin{landscape}</pre>
SystControlPlots	if set to TRUE, plots showing the shape effect of a given systematic (before and after smoothing/symmetrisation) will be produced (default: TRUE)
SystErrorBars	TRUE by default, add stat error bars to syst variations in syst plots, set to FALSE to disable
SystDataPlots	if set to TRUE, plots showing the shape effect of a given systematic (before and after smoothing/symmetrisation) on top of the nominal sum of samples will be produced. Data are then plotted in both the upper frame and the ratio.
CorrelationThreshold	Threshold used to draw the correlation matrix (only systematics with at least one correlation larger than than draw) (0.05:5%)
SignalRegionsPlot	list of regions to put in SignalRegionsPlot and PieChartPlots; use "EMPTY" to put an empty entry, "ENDL" to specify end of line. This specifies the order of regions plotted in signal region S/B plots and pie chart plots, as well as number of regions per row.
LumiLabel	label for luminosity to be put on plots
CmeLabel	label for center-of-mass energy to be put on plots
RankingMaxNP	max number of NP to show in ranking plot

Option	Function
RankingPlot	NP categories in gammas or systs, if set to Sysys(Gammas) then plot only systs(Gammas) in ranking, default produce plot for systs+gammas, can also set to all to have the 3 plots.
PlotLabel	to specify Internal, Preliminary, etc... If set to <code>none</code> the whole label will be removed, if set to <code>empty</code> , only the plot label will be added (publication style)
CleanTables	if set to TRUE, a cleaned version of the tex tables is created (basically removing the "#") - to be expanded
SystCategoryTables	if set to TRUE, additional syst tables with systematics grouped by category are created
SummaryPlotYmax	if set, it will force the summary plot to use this value as max y-axis value
SummaryPlotYmin	if set, it will force the summary plot to use this value as min y-axis value
SummaryLogY	Can be set to <code>TRUE</code> (default) or <code>FALSE</code> , controls the use of logarithmic scale on Y axis for summary plots
RatioYmax	if set, it will specify the max of the range of the ratio plots
RatioYmin	if set, it will specify the min of the range of the ratio plots
RatioYmaxPostFit	if set, it will specify the max of the range of the ratio plots, for post-fit only
RatioYminPostFit	if set, it will specify the min of the range of the ratio plots, for post-fit only

Option	Function
GetChi2	if set to TRUE (or STAT+SYST), for pre- and post-fit plots the extended chi2 test is done, and results are printed on the screen for each plot when running d and/or p; can be set to STAT (or STAT-ONLY) for stat-only chi2
SummaryPlotRegions	list of regions to be shown in summary plot (useful to specify a custom order)
DoSummaryPlot	if set to FALSE, no summary plot is created
DoMergedPlot	if set to TRUE, a merged plot of all the region groups specified with the RegionGroups option is created
DoTables	if set to FALSE, no tables are created
DoSignalRegionsPlot	if set to FALSE, no signal regions plot is created
DoPieChartPlot	if set to FALSE, no background composition pie-chart plot is created
DoSystNormalizationPlots	Set to <code>FALSE</code> to disable the normalization summary plot that is produced during the <code>w</code> step
RegionGroups	groups specified here will cause additional yield tables to be created per group, and also merged plots per group if DoMergedPlot is set to TRUE
SummaryPlotLabels	DEPRECATED - labels to be used per region group in summary plot (only if FourTopStyle is set)
SummaryPlotValidationRegions	regions to be included in validation region summary plot (default: all)

Option	Function
SummaryPlotValidationLabels	DEPRECATED - labels to be used per set of regions in validation-region summary plot (only if FourTopStyle is set)
UsePDGRounding	If set to <code>TRUE</code> will use PDG rounding to yield tables (both .txt and .tex)
UsePDGRoundingTxt	If set to <code>TRUE</code> will use PDG rounding to yield tables (only .txt)
UsePDGRoundingTex	If set to <code>TRUE</code> will use PDG rounding to yield tables (only .tex)
PrePostFitCanvasSize	Set width and height for canvas for pre/post-fit plots
POIPrecision	Integer value N, $N \geq 1$ and $N \leq 5$. Will tell the code to use N decimal places for norm factor mean value and uncertainty. Default is 2
RankingPOIName	Custom name for the POI for ranking plots. Comma separated list. The number and order must match the POI setting. Default is <code>#mu</code>
RankingPOIAxisScale	Order of magnitude of the scale for the POI axis for ranking plots. Comma separated list. The number and order must match the POI setting. Default is 1.0
RankingUpperAxisNdivision	Standard ROOT Ndivision option for upper axis for ranking plots. Comma separated list. The number and order must match the POI setting. Default is 510
SummaryCanvasSize	Set width and height for canvas for summary plots
MergeCanvasSize	Set width and height for canvas for merged plots
PieChartCanvasSize	Set width and height for canvas for pie chart plots

Option	Function
NPRankingCanvasSize	Set width and height for canvas for NP ranking plot
LabelX	Custom X position for Plot label and others on Data/MC plots.
LabelY	Custom Y position for Plot label and others on Data/MC plots.
LegendX1	Custom Legend X1 position for Plot label and others on Data/MC plots.
LegendX2	Custom Legend X2 position for Plot label and others on Data/MC plots.
LegendY	Custom Legend Y top position for Plot label and others on Data/MC plots.
LabelXSummary	Same as LabelX but for Summary plot.
LabelYSummary	Same as LabelY but for Summary plot.
LegendX1Summary	Same as LegendX1 but for Summary plot.
LegendX2Summary	Same as LegendX2 but for Summary plot.
LegendYSummary	Same as LegendY but for Summary plot.
LabelXMerge	Same as LabelX but for Merged plot.
LabelYMerge	Same as LabelY but for Merged plot.
LegendX1Merge	Same as LegendX1 but for Merged plot.
LegendX2Merge	Same as LegendX2 but for Merged plot.

Option	Function
LegendYMerge	Same as LegendY but for Merged plot.
LegendNColumns	Number of columns in Legend for Data/MC plots. Only works when PlotOptions YIELDS is NOT used.
LegendNColumnsSummary	Same as LegendNColumns but for Summary plot.
LegendNColumnsMerge	Same as LegendNColumns but for Merged plot.
ReorderNPs	If set to TRUE, fit results will show NPs and norm factors ordered according to how they appear in the config file.
RatioType	Can be used to specify what to show in the ratio pad in the Data vs. MC plots. Can be set to "DATAOVERMC" (default), "DATAOVERBKG", "SOVERB", "SOVERSQRT(B)", "SOVERSQRT(S+B)", "DATAOVERBWITHS" (i.e. showing Data/B as for "DATAOVERBKG", but with (S+B)/B overlayed).
AddAliases	List of semicolon separated <code>alias:variable</code> pairs to setup some variables as alias. Long expressions can be cumulate with this setup into shorter aliases to avoid a possible ROOT crash during the ntuple reads. This option must be used in quote e.g. <pre>"sum_pt:pt[0]+Alt\$(pt[1],0);func:MyFunc(pt,eta,phi,e)"</pre>
UnfoldingShowStatOnlyError	If set to <code>TRUE</code> (default is <code>FALSE</code>), will add stat-only error band on the unfolded plots. <i>Requires to run StatOnlyFit=TRUE</i> to get the stat only results. When Using <code>StatOnlyFit=TRUE</code> , this option defaults to <code>TRUE</code> .
UnfoldingShowUncertaintyBreakdown	If set to <code>TRUE</code> (default is <code>FALSE</code>), when running an UNFOLDING fit, this will produce a histogram

Option	Function
	visualising the grouped impact. It is produced during <code>GroupedImpact=combine</code> step.
UnfoldingUncertaintyBreakdownTotal	If set to <code>TRUE</code> (default is <code>TRUE</code>), will show the total uncertainty when <code>UnfoldingShowUncertaintyBreakdown</code> is set. When set to <code>FALSE</code> will only show total systematics instead of total (stat+syst) uncertainty.
NoPrePostFitLabel	If set to <code>TRUE</code> (default is <code>FALSE</code>), will remove the "Pre-fit" and "Post-fit" labels from the plots.
PreFitLabel	The label to be shown in the pre-fit plots. Default is "Pre-fit".
PostFitLabel	The label to be shown in the post-fit plots. Default is "Post-fit".
Blinding	
BlindingThreshold	blind bins when S/B is greater than this threshold, use the <code>BlindingType</code> option for other definitions than just S/B. Does NOT impact the fit, i.e. the data points will be used in the fit, only the output plots will be blinded.
BlindingType	how to calculate the quantity to determine blinding, options are SOVERB (for S/B), SOVERSPLUSB(for S/(S+B)), SOVERSQRTB (for S/sqrt(B)) and SOVERSQRTSPLUSB (for S/sqrt(S+B)), default is SOVERB
KeepPrefitBlindedBins	if set to <code>TRUE</code> pre-fit blinding is kept in post-fit plots
HideNP	comma-separated list of nuisance parameters to be excluded from pull plots and correlation matrix

Option	Function
BlindSRs	If set, all SRs are forced to have DataType = ASIMOV. This impacts the fit results!

Fit block settings

Option	Function
FitType	can be SPLUSB (default) or BONLY to fit under the s+b or the b-only hypothesis. Use UNFOLDING for unfolding. Use EFT for Effective Field Theory.
FitRegion	can be CRSR (default) or CRONLY to fit considering both signal and control regions in the fit, or only control regions
FitBlind	specify if real data or Asimov data should be used in the fit (TRUE = Asimov data will be used or FALSE = real data will be used). By default, fits are NOT blind.
POIAsimov	value of the parameter of interest in the AsimovDataset used in the fit. Syntax: POIAsimov: POI1@1.0,POI2@2.0,POI@-2.0
NPValues	values of the nuisance parameters used to build the Asimov. Comma-separated list of NP:value (e.g. alpha_ttbarbb_XS:1,alpha_ttbarbcc_XS:1.5). NB: if this is set, no mixed fit is performed in case of a mixture of regions with DataType=DATA and =ASIMOV (see Region->DataType option).
FixNPs	values of the nuisance parameters used to be fixed in the fit. Comma-separated list of NP:value (e.g. alpha_ttbarbb_XS:1,alpha_ttbarbcc_XS:1.5), currently only implemented for the f step

Option	Function
doLHscan	comma separated list of names of the POI or NP from which you want to produce the likelihood scan, if first element of the list is "all" then all systematics are profiled
do2DLHscan	produces 2D likelihood scan between the chosen parameters. Syntax: "paramX1,paramY1:paramX2,paramY2". Warning takes long time. You can reduce the number of steps via <code>LHscanSteps</code> . Alternatively you can split up the 2D scan in slices with <code>LHscanStep</code> and <code>LHscanStepY</code> command line options
LHscanMin	minimum value for the LH scan on x-axis (default is Norm min). This also effect the x-axis in a 2D scan
LHscanMax	maximum value for the LH scan on x-axis (default is Norm max). This also effect the x-axis in a 2D scan
LHscanSteps	number of steps on the LH scan (default is 30). Value has to be between 3 and 500. This also effect the x-axis in a 2D scan
LHscanMinY	minimum value for the 2D-LH scan on y-axis (default it Norm min)
LHscanMaxY	maximum value for the 2D-LH scan on y-axis (default is Norm max)
LHscanStepsY	number of steps on the LH scan in y-direction (default is 30, but if not specified it uses LHscanSteps)
UseMinos	comma separated list of names of the POI and/or NP for which you want to calculate the MINOS errors, if first element of the list is "all" then the MINOS errors is calculated for all systematics and POIs

Option	Function
SetRandomInitialNPval	useful to set this to >0 (e.g. 0.1) to help convergence of Asimov fits
SetRandomInitialNPvalSeed	seed used to determine initial NP settings in minimization process if SetRandomInitialNPval option is enabled
NumCPU	specify the number of CPU to use for the minimization (default = 1)
StatOnlyFit	if specified, the fit will keep fixed all the NP to the latest fit result, and the fit results will be saved with the <code>_statOnly</code> suffix (also possible to use it from command line)
DoNonProfileFit	if set to TRUE (default is FALSE), instead of the fit profiling the systematics, a set of stat-only fits will be performed, on an Asimov data-set created with one syst variation at a time
FitToys	if set to $N > 0$, N toys are generated and fitted
ToysHistoNbins	If FitToys is used, set number of bins for toys histogram output
ToysSeed	Set initial seed for the toys generation. Useful for generation of multiple independent parallel toys jobs, the outputs of which should be combined later by the user. Default is 1234. NOTE: Parallel jobs will each use a range of seeds from (ToySeed)-> (ToySeed+NToys). Each job must therefore use a unique seed starting beyond the range of the previous job - otherwise the toys will statistically overlap.
ToysStatOutput	If set to <code>TRUE</code> (default is <code>FALSE</code>) AND if <code>ToysOnlyStatFluctuation</code> is <code>TRUE</code> , the uncertainties of

Option	Function
	from the toys will be used to create the statOnly.txt file in Fits/ folder. This is useful when using unfolding with regularization
TemplateInterpolationOption	Option only for morphing, tells the code which interpolation between the templates is used. Three possible options are available: LINEAR(default)/SMOOTHLINEAR/SQUAREROOT. All of these options basically use linear interpolation but SMOOTHLINEAR approximates it by integral of hyperbolic tangent and SQUAREROOT approximates it by $\sqrt{x^2 + \epsilon}$ to achieve smooth transitions (first derivative) between the templates
BlindedParameters	A comma separated list of POI/NPs that will be written as a hexadecimal number so it is not easy to read to not accidentally unblind. When at least one parameter is set the console output of the minimization is removed.
NonProfileFitSystThreshold	When performing a NonProfileFit, systematics are not added to total if smaller than this threshold
NPValuesFromFitResultsFile	If set to a valid path pointing to a fit-result ROOT file, the NPValues for Asimov-data creation will be read from it
FitResultsRootFile	Path to the ROOT file with the post-fit results to be used for plotting/tables. Can also be passed as a command line option
InjectGlobalObservables	If set to TRUE (default is FALSE), and if NPValues or NPValuesFromFitResults are set, also the global observables are shifted in the Likelihood according to the parameter values
FitStrategy	Set Minuit2 fitting strategy, can be: 0, 1, 2 or 3. If negative value is set the default is used (1)

Option	Function
BinnedLikelihoodOptimization	Can be set to TRUE or FALSE (default). If set to TRUE, will use the <code>BinnedLikelihood</code> optimisation of RooFit that has significant speed improvements, but results in less stable correlation matrix computation
UsePOISinRanking	If set to <code>TRUE</code> (default is <code>FALSE</code>) will include POIs as NPs in the ranking
ShiftGlobalObservablesInRanking	If set to <code>TRUE</code> (default is <code>FALSE</code>) will run ranking by shifting the global observables instead of shifting and fixing a nuisance parameter
UseHesseBeforeMigrad	If set to <code>TRUE</code> (default is <code>FALSE</code>) will run hesse() method before migrad, this can help with convergence in some cases
UseNLLwithoutOffsetInLHscan	If set to <code>TRUE</code> (default) will use NLL values that don't use offset subtraction in the internal Likelihood object. Quoting from the ROOT documentation: "(if set to true) Offset likelihood by initial value (so that starting value of FCN in minuit is zero). This can improve numeric stability in simultaneous fits with components with large likelihood values"
DataWeighted	If set to <code>TRUE</code> (default is <code>FALSE</code>), the code will modify all histograms (data and prediction) by scaling them bin-wise by $N_{\text{Data}}/\text{uncertainty_Data}^2$. This allows to use the current model (likelihood) even when weighted data are used (they do not follow Poisson distribution if they are weighted) as the uncertainty of the modified data histograms is $\sqrt{N}$. Only the histograms entering the fit are affected. Note that this is valid only in the Gaussian approximation (approximately > 10 events in each bin).
RegularizationType	Only for Unfolding fit. Can be set to <code>0</code> (default, bin-by-bin constraint terms), <code>1</code> (discretized second

Option	Function
	derivative constraint), 2 for the fourth derivative, 3 for the sixth derivative or 4 for eighth derivative; it is effective only if <code>Tau</code> is specified as well, otherwise no regularization is applied
MultiStageFit	If set to <code>TRUE</code> (default is <code>FALSE</code>), the fit will be run first with a reduced set of parameters, all gammas as well as some "small" NPs will be removed (see <code>NPCutOff</code>). Then the result of this fit will be fed into another fit with all parameters. This can speed up the fit for very complex workspaces as the fit with all parameters would not have to iterate that much.
NPCutOff	Specify a fraction of NPs to be excluded from the first stage fit if the 'MultiStageFit' option is used. The NPs are ordered based on their largest impact on shape/normalisation. Note that the list of the NPs is created when running the pruning e.g. during the <code>w</code> step. The file is called <code>NPorder.txt</code> and it is used then in the fit to decide which NPs to fix.
UseRegularBinning	When set to <code>TRUE</code> (default is <code>FALSE</code>), the workspace (and the fit) will use the regular binning by modifying the input binning. This is needed for ROOT versions version < 6.26 as non-uniform binning was not supported in the earlier versions.
ErrorSigma	Value (in sigmas) used in the fit as an uncertainty on the parameters. Set to 1 by default.
UseHesse	Can be set to <code>TRUE</code> (default) or <code>FALSE</code> . Tells the code to use <code>hesse()</code> call in the minimisation during the <code>f</code> step
MaximumNumberFCNcalls	Maximum number of function calls during minimisation. If set to negative value (default) 500 x N will be used for N < 100 (RooFit default) and 5 x

Option	Function
	N^2 for $N > 100$, where N is the number of the free parameters in the fit.
ToleranceScale	Value to be used to scale the default tolerance (set to 0.01) for the fit minimisation. The default is 1, i.e. the default tolerance is used.
ToysNpValuesFile	Path to the ROOT file with the results. If set, when running toys in the <code>f</code> step, the code will read the fitted values and run the toys around the fitted values (the global observables will be shifted to the values from the txt file).
ErrorDecomposition	If set to <code>TRUE</code> (default) total error decomposition into stat and syst is printed, and NP-by-NP breakdown is written to file, for each POI. This decomposition is calculated based on the covariance matrix from the fit. The full breakdown of impacts is saved to txt file (one per POI, inside the <code>Fits/</code> directory) with the format "component symmetrized-impact impact-up impact-down"
UseAutoDiff	If set to <code>TRUE</code> (default is <code>FALSE</code>) the likelihood minimisation will use <code>clad</code> to calculate the derivatives analytically for the gradient. All HistFactory models are supported, also some extensions e.g. using simple polynomials for reparametrisation (such as EFT fit)
FitToysRandomInitialNormFactorValues	Allows to set random initial values for the NormFactors during the toys fitting (in the <code>f</code> step). This might be needed for fits with multiple minima, e.g. EFT fit. Syntax: comma separated list of <code>normFactorName:minimum:maximum</code> , where <code>normFactorName</code> is the name of the NF to be randomised, <code>minimum</code> and <code>maximum</code> are the values to be used in the random uniform generation for the starting points. E.g. <code>ctg:-1:1,ctW:-0.5:0.5</code>

Option	Function
FitToysLHscanCondition	Allows to run LH scan for a toy dataset falling into some range of the fitted values during the toys fitting (in the <code>f</code> step). Syntax: comma separated list of <code>normFactorName:minimum:maximum</code> , where <code>normFactorName</code> is the name of the NF to be randomised, <code>minimum</code> and <code>maximum</code> are the values where the given toys should fall. Will run the LH scan only on the first "hit". E.g. <code>ctg:-1:1,ctW:-0.5:0.5</code>
AddNFsToPOIs	If set to <code>TRUE</code> (default is <code>FALSE</code>), will add all NormFactors as parameters of interest. Can also be set via command line option
ToysLHScanForAll	If set to <code>TRUE</code> (default is <code>FALSE</code>), will run LHscan for all POIs for all toys (this can be very slow!). No plots will be produced, but the results will be stored in a ROOT file in the <code>LHoodPlots/</code> folder
ToysOnlyStatFluctuation	If set to <code>TRUE</code> (default is <code>FALSE</code>), when generating the toys, only the statistical fluctuations will be considered and not the systematic uncertainties. The fit for each toy will use the full model. To run StatOnly toys with StatOnly fits, generate StatOnly workspace and run the toys
NLLOffset	Set the likelihood offsetting mode. Allowed values are: <code>INITIAL</code> (default), which offsets the likelihood by its initial value, <code>BIN</code> , which applies a bin-by-bin offset with a template histogram model based on the observed data, or <code>NONE</code> . Refer to <code>RooFit::RooAbsPdf::createNLL()</code> documentation for more information.

`Limit` block settings

Option	Function
LimitType	can be ASYMPTOTIC (default) or TOYS (uses WS as in the 'f' step)
POI	Specifies which POI to use for the limit. If nothing is set, the first POI defined under Job will be used
LimitBlind	can be TRUE or FALSE (TRUE means that ALL regions are blinded). Use <code>LimitBlind: False</code> for mixed data/MC fits for the dataset to be used in the limit estimate
SignalInjection	if set to TRUE, expected signal with signal injection is evaluated
SignalInjectionValue	Value for the injected signal
ParamName	Name for the parameter in the output ROOT file
ParamValue	Value of the parameter in the output file (e.g. 172.5 for top mass)
OutputPrefixName	Prefix for the output ROOT file
ConfidenceLevel	Confidence level for the CLs. Default is 0.95
SplusBToys	Only for limit type <code>TOYS</code> . Set the number of toys for SplusB
BonlyToys	Only for limit type <code>TOYS</code> . Set the number of toys for Bonly
ScanSteps	Only for limit type <code>TOYS</code> . Number of steps for limit scanning
ScanMin	Only for limit type <code>TOYS</code> . Min for limit scanning
ScanMax	Only for limit type <code>TOYS</code> . Max for limit scanning
ToysSeed	Only for limit type <code>TOYS</code> . Set initial seed for the toys generation. Useful for generation of multiple independent parallel toys jobs, the outputs of which should be combined later by the user. Default is 1234

Option	Function
LimitPlot	Only for limit type <code>TOYS</code> . If set to <code>TRUE</code> (default) will produce brazilian-style plot in the Limits/ folder
LimitFile	Only for limit type <code>TOYS</code> . If set to <code>TRUE</code> (default) will produce ROOT file in the Limits/ folder with information per point
FitStrategy	Set Minuit2 Hesse strategy for limit extraction, can be: 0, 1, 2 or 3. The default is 1 (No <code>Hesse</code>). Accuracy and computational cost increase with this value.
ToysUsexRooFit	Only for limit type <code>TOYS</code> . Use the xRooFit backend to compute the toy-based limits. Valid option are FALSE (default), TRUE, and AUTO. A value of TRUE uses the specified <code>SplusBToys</code> , <code>BonlyToys</code> , <code>ScanSteps</code> , <code>ScanMin</code> and <code>ScanMax</code> values to configure the scan, while a value of AUTO uses xRooFit's auto-configuration mode.
UseAutoDiff	If set to <code>TRUE</code> (default is <code>FALSE</code>) the likelihood minimisation will use <code>clad</code> to calculate the derivatives analytically for the gradient. All HistFactory models are supported, also some extensions e.g. using simple polynomials for reparametrisation (such as EFT fit)

Significance block settings

Option	Function
SignificanceType	can be ASYMPTOTIC (default) or TOYS (uses WS as in the <code>f</code> step).
SignificanceBlind	can be TRUE or FALSE (TRUE means that ALL regions are blinded). Use <code>SignificanceBlind: False</code> for mixed data/MC fits for the dataset to be used in the limit estimate
POI	Specifies which POI to use for the significance. If nothing is set, the first POI defined under Job will be used

Option	Function
POIASimov	value of the POI to inject in the Asimov dataset in SignificanceBlind is set to TRUE
ParamName	Name for the parameter in the output ROOT file
ParamValue	Value of the parameter in the output file (e.g. 172.5 for top mass)
OutputPrefixName	Prefix for the output ROOT file
SplusBToys	Only for significance type <code>TOYS</code> . Set the number of toys for SplusB. Default value is 0 as this is not needed for significance, it is only useful for plots for comparison
BonlyToys	Only for significance type <code>TOYS</code> . Set the number of background-only toys
ToysSeed	Seed for the random numbers in case of TOYS type
SignificancePlot	If set to FALSE (default is TRUE) will disable plotting of the significance plot in case of TOYS type
ToysUsexRooFit	Only for limit type <code>TOYS</code> . Use the xRooFit backend to compute the toy-based limits. Valid option are FALSE (default) or TRUE
FitStrategy	Set Minuit2 fitting strategy, can be: 0, 1, 2 or 3. If negative value is set the default is used (1)
UseAutoDiff	If set to <code>TRUE</code> (default is <code>FALSE</code>) the likelihood minimisation will use <code>clad</code> to calculate the derivatives analytically for the gradient. All HistFactory models are supported, also some extensions e.g. using simple polynominals for reparametrisation (such as EFT fit)

Options block settings

additional options, accepting only float as arguments - useful for adding your functionalities & flags in a quick way, since they need minimal changes in the code ...

Option	Function
LogSignalRegionPlot	user-defined variable to produce plots with log scaled y-axis. Default is 0.0 (no)
LogXSignalRegionPlot	user-defined variable to produce plots with log scaled x-axis. Default is 0.0 (no)
MergeLogY	Relevant if <code>DoMergedPlot</code> is TRUE: if set to > 0, y axis will be log in merged plot
MergeLogYmin	Relevant if <code>DoMergedPlot</code> is TRUE: to set the minimum of the y axis
MergeUniqueY	Relevant if <code>DoMergedPlot</code> is TRUE: if set to > 0, each section of the merged plot will have the same y axis - otherwise each plot is rescaled to fit the canvas
MergeYfrac	Relevant if <code>DoMergedPlot</code> is TRUE: to be used to adjust the vertical position of the region labels
MergeYmaxScale	Relevant if <code>DoMergedPlot</code> is TRUE: to be used to adjust the maximum y
MergeFixLastBinWidth	Relevant if <code>DoMergedPlot</code> is TRUE: if set to > 0, the last bin of each region will have the same size as the next-to-last
MergeRegionLabelAlign	Relevant if <code>DoMergedPlot</code> is TRUE: alignment for the region labels in each section (1=left, 2=center, 3=right - default = center)
CanvasHeight	Height of the canvas for the pre/post fit plots
CanvasWidth	Width of the canvas for the pre/post fit plots
CanvasWidthMerge	Width of the canvas for the merged plot
CanvasWidthSummary	Width of the canvas for the summary plot
LegendNColumns	Number of column for the legend

Option	Function
NormSigSRonly	Normalise signal in the SR only for plotting
PieChartSize	Self-explanatory
PoissonizeData	Set data to random poisson distributed values per bin
RandomizeMC	Set MC to random gaussian distributed values per bin
PrefitRatioMax	Set maximum for the prefit ratio Y axis
ReduceRanking	Shift the ranking NPs by a product of the standard shift and the provided value
SignalRegionSize	Controls canvas size for the <code>DoSignalRegionsPlots</code> plots
SystFillStyle	Controls error band fill style
SystPlotRatioRange	Controls the ratio Y axis range for the systematic "red-blue" plots

Region block settings

Option	Function
VariableTitle	it's the label which will be displayed on the x-axis in the plots
Label	it's the label which will be showed on the plots and specifies which region is shown, if 'NONE' is set, no label will be shown
ResponseMatrixFile(s)	same as for Job but for Region
ResponseMatrixFileSuff(s)	same as for Job but for Region
ResponseMatrixName(s)	same as for Job but for Region

Option	Function
ResponseMatrixNameSuff(s)	same as for Job but for Region
ResponseMatrixPath(s)	same as for Job but for Region
ResponseMatrixPathSuff(s)	same as for Job but for Region
AcceptanceFile(s)	same as for Job but for Region
AcceptanceFileSuff(s)	same as for Job but for Region
AcceptanceName(s)	same as for Job but for Region
AcceptanceNameSuff(s)	same as for Job but for Region
AcceptancePath(s)	same as for Job but for Region
AcceptancePathSuff(s)	same as for Job but for Region
SelectionEffFile(s)	same as for Job but for Region
SelectionEffFileSuff(s)	same as for Job but for Region
SelectionEffName(s)	same as for Job but for Region
SelectionEffNameSuff(s)	same as for Job but for Region
SelectionEffPath(s)	same as for Job but for Region
SelectionEffPathSuff(s)	same as for Job but for Region
MigrationFile(s)	same as for Job but for Region
MigrationFileSuff(s)	same as for Job but for Region
MigrationName(s)	same as for Job but for Region

Option	Function
MigrationNameSuff(s)	same as for Job but for Region
MigrationPath(s)	same as for Job but for Region
MigrationPathSuff(s)	same as for Job but for Region
NormalizeMigrationMatrix	Whether to normalize the migration matrix. Default is <code>TRUE</code>
TexLabel	label for tex files
ShortLabel	same as above, but a shorter version for plots with smaller available place
LumiLabel	label for luminosity to be put on plots
CmeLabel	label for center-of-mass energy to be put on plots
LogScale	set it to <code>TRUE</code> to have log-scale when plotting this region
LogScaleX	set it to <code>TRUE</code> to have log-scale on the X axis when plotting this region
HistoFile(s)	only for option HIST, the file name (or names, comma-separated) to be used
HistoFileSuff(s)	only for option HIST, the file name suffix (or names, comma-separated) to be used
HistoName(s)	only for option HIST, the histogram name (or names, comma-separated) to be used
HistoNameSuff(s)	only for option HIST, the histogram name suffix (or names, comma-separated) to be used

Option	Function
HistoPath(s)	only for option HIST, the path (or path, comma-separated) where to find the histogram files for this region
HistoPathSuff(s)	only for option HIST, the path suffix (or suffixes, comma-separated) where to find the histogram files for this region
Variable	only for option NTUP, the variable (or expression) inside the ntuple to plot can define a variable as X
VariableForSample	only for option NTUP, allows to set exceptions for Variable. This is a very useful feature when using TRF only in some samples. Comma-separated list of sample:variable (e.g. wjets:met_met/1e3,zjets:Mbbb/1e).
Selection	only for option NTUP, the selection done on the ntuple for this region
SelectionForSample	only for option NTUP, sample specific selection. Format: sample1:selection1,sample2:selection2
NtupleName(s)	only for option NTUP, the name (or names, comma-separated) of the tree for this region
NtupleNameSuff(s)	only for option NTUP, the name suffixes (or names, comma-separated) of the tree for this region
NtupleFile(s)	only for option NTUP, the file (or files, comma-separated) of the tree for this region
NtupleFileSuff(s)	only for option NTUP, the file suffixes (or files, comma-separated) of the tree for this region
NtuplePath(s)	only for option NTUP, the path (or paths, comma-separated) where to find the ntuple files for this region

Option	Function
NtuplePathSuff(s)	only for option NTUP, the path suffix (or suffixes, comma-separated) where to find the ntuple files for this region
MCweight	only for option NTUP, the additional weight used in this region (for MC samples only)
Rebin	if specified, the histograms will be rebinned merging N bins together, where N is the argument (int)
Binning	if specified, the histograms will be binned according to the new binning specified, in the form of <code>0,10,20,50,100</code> . The parameter <code>AutoBin</code> allows to use algorithms to define the binning. Example - <code>Binning: "AutoBin","TransfoD",5.,6.</code> (<code>TransfoF</code> also available, <code>5.</code> and <code>6.</code> are parameters of the transformation; <code>TransfoJ</code> takes three parameters). A popular setting is <code>TransfoD</code> with both parameters set to the same value; the amount of bins used is equal to the sum of both parameters. If used in background region and <code>zSig!=0</code> (first parameter, <code>=0</code> gives flat background) then need a comma separated list of backgrounds to use instead of signal to compute the binning. More information about the automatic binning algorithms is found in the FAQ .
Rebinning	if specified, the histograms will be rebinned according to the new binning specified, in the form of <code>0,10,20,50,100</code> . Differently from the Binning option, this one performs the rebinning after the original histograms are created. This means that this option can be changed (or removed) and then rerun the <code>b</code> step without the need to rerun the <code>h</code> or <code>n</code> steps (i.e. standard TH1::Rebin method).
BinWidth	if specified, two things are done: this number is used to decorate the y axis label and the bin content is scaled for bins with a bin width different from this number. The y axis label also includes as unit anything specified in square brackets in <code>VariableTitle</code> , e.g. <code>"GeV"</code> when the title is <code>"jet pT [GeV]"</code> .

Option	Function
BinLabels	if specified, bin labels are set according to provided comma separated list (list length must be equal to number of bins)
BinDividers	if specified, vertical lines are drawn at low edges of bins set according to provided comma separated list
RangeLabels	if specified under the form like "Region 1", "Region 2" and BinDividers is set, labels will be drawn at the first bin and those defined by BinDividers
BinTickMarks	if specified under the form like 1, 9, 17, 19, tick marks on the X-axis will be put only at the low edges of the specified bin numbers; the minimal value is 1, the maximal value is Nbins+1, the list should be sorted
BinTickMarksLabels	if specified under the form like "0.29", "1.37", "", "3.46", the customised X-axis tick marks setup using the BinTickMarks option will have labels set using this list of strings; the list should be of the same size as BinTickMarks
Type	can be SIGNAL, CONTROL or VALIDATION; used depending on Fit->FitType; if VALIDATION is set, the region is never fitted; default is SIGNAL
DataType	ASIMOV or DATA. Is ASIMOV is set in some of the regions and DATA in some other ones, limits, significance and final fit are computed without taking into account the data in this region, but creating a modified Asimov data-set through a projection of the fit performed in the regions with DATA only. NB: this "Mixed fit" feature only works if no NPValues option is set (see Fit->NPValues).
Ymax	if set, it will force the plot to use this value as max y-axis value
Ymin	if set, it will force the plot to use this value as min y-axis value

Option	Function
RatioYmax	if set, it will specify the max of the range of the ratio plot for this region only
RatioYmin	if set, it will specify the min of the range of the ratio plot for this region only
RatioYmaxPostFit	if set, it will specify the max of the range of the ratio plot for this region only, for post-fit only
RatioYminPostFit	if set, it will specify the min of the range of the ratio plot for this region only, for post-fit only
AutomaticDropBins	Set this to TRUE to automatically drop bins above the <code>BlindingThreshold</code> from the fit. Default setting is FALSE, and use of <code>DropBins</code> also sets this option to FALSE. When FALSE, the bins will only be visually blinded if specified in <code>BlindBins</code> , and they will enter the fit. Make sure to set <code>KeepPrefitBlindedBins</code> to TRUE to also have post-fit plot blinded.
DropBins	A comma-separated list of bins (bin indices, <i>not</i> bin edges) which will be dropped from the fit. A copy of the original histogram will be created without the dropped bins, which will be used for the fit. The bins will <i>not</i> be visually blinded, unless specified in <code>BlindBins</code> .
BlindBins	A comma-separated list of bins (bin indices, <i>not</i> bin edges) which will be visually blinded. The bins will enter the fit, unless dropped by <code>DropBins</code> or <code>AutomaticDropBins</code> .
Group	if specified, regions of the same group appear together in several places, see <code>RegionGroups</code> option
YaxisTitle	title of y-axis used for plots of the region
YmaxScale	scales range of y-axis (default: 2.0, meaning the maximum axis value is twice the largest yield in any bin)

Option	Function
Ymax	maximum value on y-axis
SkipSmoothing	if smoothing of nominal samples is used, this option can be used to disable smoothing per region (default: FALSE)
XaxisRange	Manually call 'SetRangeUser()' on X axis. Needs two parameters(floats): min,max
NumberOfRecoBins	Number of reco bins in this region when Unfolding is used. If set, it will be used for sanity checks, but it does not have to be set
IsBinOfRegion	can be set in order to declare this region as corresponding to a certain bin of another region (to be set as VALIDATION region and to be declared previously in the config file); in this way, one can perform a two-dimensional fit, by keeping the systematic smoothing in both directions: for each systematic, the smoothing applied to the VALIDATION region indicated will be propagated to the overall part of the systematic variation in this region; syntax: "region-name":binNmbr (NB: bin numbering here starts at 1 due to technical reasons)
RandomNFscale	Can be set to generate a random number that will be used instead of the nominal value when making the pre-fit plots. This can be used to blind the prefit plots to the given POI. Syntax is: "nameOfThePoi1:min:max,nameOfThePoi2:min:max", e.g. RandomNFscale: topWidth:1.8:1.9, ttbarXsec:0.95:1.05
DrawDataMinusBkg	Can be set to NONE (default), ERRONSIGNAL or ERRONBKG . When set to ERRONSIGNAL or ERRONBKG , the code will produce a data - background plot comparing the signal only prediction with the background-only total uncertainty band. For ERRONSIGNAL the error band is centered around the signal, while for ERRONBKG the error band is centered around 0 (background - background).
PruningPlotSamples	Comma separated list of samples to be used for the pruning plot for the given region. By default, all samples are used.

Option	Function
TransfoErr	Using a given error of bin to implement a new bin in transformation. Default set to 10 (not used). Redefine default in TransformationF to 0.1 that was hardcoded.
PlotSubdir	Subdirectory of the main plot directory to be used for output plots for this region. If unset, plots are written to the main directory.
FriendPath(s)	Path to the ntuples for the friend TTree
FriendPathSuff(s)	Path suffices to the ntuples for the friend TTree

Sample block settings

Option	Function
Type	can be SIGNAL, BACKGROUND, DATA, GHOST; default is BACKGROUND; GHOST means: no syst, not drawn, not propagated to workspace
Title	title shown on the legends
TexTitle	title shown on tex tables
Group	if specified, sample will be grouped with other samples with same group and this label will be used in plots
HistoName(s)	valid only for option HIST; name(s) of histogram to read
HistoFile(s)	valid only for option HIST; which root file(s) to read (excluding the suffix ".root"); this will be combined with Fit->HistoPath to build the full path

Option	Function
HistoPath(s)	valid only for option HIST; it's the path(s) where the input root files containing the histograms are stored
HistoNameSuff(s)	valid only for option HIST; suffix(es) for the name of histogram(s) to read
HistoFileSuff(s)	valid only for option HIST; suffix(es) for the file name of histogram(s) to read
HistoPathSuff(s)	valid only for option HIST; suffix(es) for the path of histogram(s) to read
HistoFolderName	valid only for option HIST; Name of the folder for the histogram path inside the ROOT file. This can be used e.g. if the Region setting has HistoName of a form of <code>Folder/path</code> where <code>Folder</code> will be replaces with the provided setting
NtupleName(s)	valid only for option NTUP; name(s) of tree to read
NtupleFile(s)	valid only for option NTUP; it's the file name(s) where the input ntuples are stored. UNIX-style globbing supported on UNIX-systems (not Windows)
NtuplePath(s)	valid only for option NTUP; it's the path(s) where the input root files containing the ntuples are stored. UNIX-style globbing supported on UNIX-systems (not Windows)
NtupleNameSuff(s)	valid only for option NTUP; suffix(es) for the name of tree to read
NtupleFileSuff(s)	valid only for option NTUP; suffix(es) for the file name(s) of tree to read
NtuplePathSuff(s)	valid only for option NTUP; suffix(es) for the path(s) of tree to read

Option	Function
FillColor	histogram fill color (not valid for data)
FillColorRGB	histogram fill color in RGB (not valid for data). This expects a triplet of RGB values between 0 and 255, e.g. <code>255, 0, 0</code> . If set, the FillColor option is ignored.
FillLineColorHEX	histogram fill and line color using the hexadecimal string
LineColor	histogram line color
LineColorRGB	histogram line color in RGB. This expects a triplet of RGB values between 0 and 255, e.g. <code>255, 0, 0</code> . If set, the LineColor option is ignored.
LineStyle	histogram line style used for signal overlays and ratios involving signals
LineWidth	histogram line width used for signal overlays
NormFactor	NormalisationFactor (free parameter in the fit); in the format <code>\<name>,nominal,min,max</code>
ShapeFactor	ShapeFactor added
NormalizedByTheory	set it to FALSE for data-driven backgrounds (MCweight, Lumi and LumiScale from Job and Region will be ignored)
MCweight	only for option NTUP, the additional weight used in this sample
Selection	valid only for option NTUP; additional selection for this region
Regions	set this to have the sample only in some regions
Exclude	set this to exclude the sample in some regions

Option	Function
LumiScale(s)	set this to scale the sample by a number; if more numbers are set, use a different one for each file / name / path...
IgnoreSelection	if set to <code>TRUE</code> , selection from Job and Region will be ignored. If set to a string, that particular substring will be ignored from the selection string
IgnoreWeight	if set to <code>TRUE</code> , weights from Job and Region will be ignored. If set to a string, that particular substring will be ignored from the weight string
UseMCstat	if set to <code>FALSE</code> , makes the fitter ignore the stat uncertainty for this sample
UseSystematics	has to be set to <code>TRUE</code> to allow systematics on the GHOST samples
MultiplyBy	if specified, each sample hist is multiplied bin-by-bin by another sample hist, in each of the regions
DivideBy	if specified, each sample hist is divided bin-by-bin by another sample hist, in each of the regions
AddSample(s)	if specified, each sample hist gets added bin-by-bin another sample hist, in each of the regions
SubtractSample(s)	if specified, each sample hist gets subtracted bin-by-bin another sample hist, in each of the regions
NormToSample	normalize yield of sample to yield of the sample specified by name, normalization happens per region
Smooth	if set to <code>TRUE</code> , the nominal histograms are smoothed (based on <code>TH1::Smooth</code> but taking into account the original stat uncertainty)

Option	Function
AsimovReplacementFor	only for GHOST and EFT samples; if set, the creation of custom Asimov data-set(s) is triggered; It is possible to create multiple different Asimovreplacements from this one file the syntax via setting multiple asimovs seperated by a comma. Use as: 'AsimovReplacementFor: "dataset1=sample1","dataset2=sample1","dataset3=sample1"', where "dataset" is the name of a custom Asimov dataset one wants to create (the same name will have to be set under Job->CustomAsimov in order to use it) and "sample" is the sample this GHOST sample will supersede.
SeparateGammas	if set to TRUE, the sample will not contribute to the overall gamma factors for MC stat, but a separate set of them will be added for this sample (through the SHAPE systematic technology); NB: you need to re-run the <code>n</code> or <code>h</code> step if you want to decorrelate the gammas on existing inputs (<code>wf</code> is not enough). If <code>MCstatThreshold</code> is set to <code>NONE</code>, this option is ignored
CorrelateGammasInRegions	to be used only together with SeparateGammas; can be used to correlate MC stat across regions; example: "SR1:SR2,CR1:CR2:CR3" will use the same NP for the MC stat in each bin of SR1 and SR2 and in each bin of CR1, CR2 and CR3
CorrelateGammasWithSample	to be used only together with SeparateGammas; can be used to correlate MC stat of this sample with those of another sample (example usecase: when one sample is derived from another one through reweighting, and they are both used in the fit)
Morphing	add this to each template you have, to do a template fit / morphing; syntax is <code><name-of-parameter>,<value-corresponding-to-this-template></code>; the POI should be set to <code><name-of-parameter></code>

Option	Function
MCstatScale	scales up/down the MC stat size; useful to project sensitivity to larger MC samples
SystFromSample	set it to TRUE (default FALSE) to have this sample inheriting all the systematics from another sample
UseGaussianShapeSysConstraint	set it to TRUE (default FALSE) will use Gaussian constraint term for the SHAPE systematics for this sample, e.g. when SeparateGammas: TRUE is used
NoPruning	set it to TRUE (default FALSE) will not apply pruning for the sample
GridDID	Rucio DIDs for ServiceX to deliver minimal ROOT ntuples from grid; More information is found in the ServiceX for TRexFitter .
Template	Tell the code that this is a template used for template morphing. Syntax 1D: <code>target@parameterName:value</code> (example: <code>Signal_1p5_172p5@topMass:172.5</code>), syntax 2D: <code>target@parameterName1:value1,parameterName2:value2</code> (example: <code>Signal_1p5_172p5@topMass:172.5,topWidth:1.32</code>). Only one SIGNAL sample per target can have this, other samples have to be of type GHOST
AlternativeRefInTemplateMorphing	Provide the name of the alternative sample (e.g. fast simulation) for the calculation of the morphing parameterisation
UseFriend	Tells the code to use the friend TTree functionality (default is <code>FALSE</code>)
FriendPath(s)	Path to the ntuples for the friend TTree
FriendPathSuff(s)	Path suffices to the ntuples for the friend TTree

Option	Function
EFTParam	Parameters for the EFT sample, e.g. <code>ctW + ctW^2</code>
EFTSMReference	SM reference sample for the EFT fit. For the per-bin normalisation processing, set to "NONE" to identify the SM reference.
EFTTitle	Title for the EFT parameter, e.g. "ctW=c_{tW}"
EFTValue	Value for the parameter for the EFT, e.g. <code>ctW=-3.0</code>

`NormFactor` block settings

Option	Function
Samples	comma-separated list of samples on which to apply the norm factor, or <code>NONE</code> to not apply it to any (useful when using expressions of normalization factors); to include samples used for unfolding, can directly include names of <code>UnfoldingSample</code> s and/or names of sub-samples created internally (wildcards can be used)
Regions	comma-separated list of regions where to apply the norm factor
Exclude	comma-separated list of samples/regions to exclude
Title	title of the norm factor
TexTitle	title of the norm factor for latex
Nominal	nominal value
Min	min value
Max	max value

Option	Function
Constant	set to TRUE to have a fixed norm factor
Category	major category to which the NormFactor belongs (instrumental, theory, ttbar, ...)
SubCategory	minor category for the NormFactor, used to evaluate impact on POI per SubCategory in "i" step, defaults to "NormFactors", do not use "Gammas", "FullSyst", or "combine" as SubCategory names (reserved for special functionality)
Expression	a way to correlate this norm factor with other norm factors (using AddPreprocessFunction); two arguments, in the form <code><expression>: <dependencies></code>, where <code><dependencies></code> should contain the names of the norm factors the expression depends on, their nominal values and existence ranges [example: <code>(1.+Pmag*cos(theta))/2.:Pmag[0.9,0,1],theta[0,0,3.14]</code>]; in case the dependencies correspond to exisisting NormFactors, the nominal, min and max values will be in any case taken from the existing NormFactors; in this case, it is possible to avoid specifying the range(s) [example: <code>(1.+Pmag*cos(theta))/2.:Pmag,theta</code>] but in this case, it needs to be done for all the parameters in the expression
Tau	If set, a constraint term will be added to the likelihood for this NF, in the same way as for the Tikhonov regularization (used in unfolding); the constraint will be of the form $\exp(-(\frac{1}{2}) * \tau^2 * (NF-\mu)^2)$, where μ is the Nominal value for the NF, so the larger is tau, the stronger the constraint
CorrelateUnfolding	Only for Unfolding fit. If a name of the Unfolding objects is provided. It will correlate (replace) this NF for a total cross-section for the selection object. Can be used e.g. for signal events in a CR.

ShapeFactor block settings

Option	Function
Samples	comma-separated list of samples on which to apply the shape factor
Regions	comma-separated list of regions where to apply the shape factor
Exclude	comma-separated list of samples/regions to exclude
Title	title of the shape factor
TexTitle	title of the shape factor for latex
Nominal	nominal value (for all bins)
Min	min value (for all bins)
Max	max value (for all bins)
Constant	set to TRUE to have a fixed shape factor
Category	major category to which the ShapeFactor belongs (instrumental, theory, ttbar, ...)
Expression	Allows to reparametrise the ShapeFactors. Syntax: "formula:dependencies" like for NormFactor, separated with @ . The order has to match the order of bins in the region. Example for a region with 3 bins: <pre>(1+2X+Y):X[1,0,2],Y[1,0,2]@(1+X+4Y):X[1,0,2],Y[1,0,2]@(1+X+Y):X[1,0,2],Y[1,0,2]</pre> The new parameters are automatically added to the NormFactors and POIs. Exactly one Sample and one Region has to be provided for this ShapeFactor.

Systematic block settings

Option	Function
Samples	comma-separated list of samples on which to apply the systematic

Option	Function
Regions	comma-separated list of regions where to apply the systematic
Exclude	comma-separated list of samples/regions to exclude
ExcludeRegionSample	comma-separated list of region:sample to exclude
Type	can be HISTO, OVERALL, SHAPE (this refers to the HistFactory Shape Systematic, i.e. uncorrelated bin-by-bin) or STAT (this refers to auto-creation of one systematic from stat uncertainty for each bin of corresponding region - DEPRECATED)
Title	title of the systematic (will be shown in plots)
TexTitle	title of the systematic in the latex files
StoredName	if specified, will be used to read and write histograms in the root files under Histograms/ instead of the syst name; useful to decorrelate without re-creating histograms
NuisanceParameter	if specified, this will be given to RooStats instead of the syst name; useful (and recommended) way to correlate systematics
IsFreeParameter	if set to TRUE, the associated constraint term is uniform instead of Gaussian (use with caution), requires workspace re-creation (via <code>w</code>), NOTE: this currently does NOT work correctly with shape variations (HistoSys modifiers), see bug report
IsCorrelated	if set to TRUE, the systematic is considered correlated to nominal, currently only has effect on the BootstrapSample option. TRUE by default.

Option	Function
Category	major category to which the systematic belongs (instrumental, theory, ttbar, ...): used to split pulls plot for same category
SubCategory	minor category for the systematic, used to evaluate impact on POI per SubCategory in "i" step, defaults to Category setting if it is used, otherwise defaults to "Uncategorised", do not use "Gammas", "FullSyst", or "combine" as SubCategory names (reserved for special functionality)
CombineName	A unique string for each systematic that you want to combine into a single systematic (e.g.) envelope. This needs to be set for every systematic that needs to be combined. The code will then combine all systematic and modify the <i>first one</i> and then set all other systematics manually to zero. This is executed during the <i>b</i> step.
CombineType	Can be <code>ENVELOPE</code> , <code>STANDARDDEVIATION</code> (divide by <code>N-1</code>), <code>STANDARDDEVIATIONNODDOF</code> (no delta degree of freedom, i.e. divide by <code>N</code>), <code>HESSIAN</code> (divide by <code>1</code>) or <code>SUMINSQUARES</code> . Tells the code how to combine the systematics with the same <code>CombineName</code> . <code>STANDARDDEVIATION</code> , <code>STANDARDDEVIATIONNODDOF</code> , and <code>HESSIAN</code> do OneSided symmetrisation while for <code>ENVELOPE</code> it is recommended to use the symmetrisation for the individual components. <code>SUMINSQUARES</code> option will sum systematic uncertainties in squares (bin-by-bin)
HistoPathUp	only for option HIST, for HISTO or SHAPE systematic: histogram file path for systematic up variation
HistoPathDown	only for option HIST, for HISTO or SHAPE systematic: histogram file path for systematic down variation

Option	Function
HistoPathSufUp	only for option HIST, for HISTO or SHAPE systematic: suffix of the histogram file names for systematic up variation
HistoPathSufDown	only for option HIST, for HISTO or SHAPE systematic: suffix of the histogram file names for systematic down variation
HistoFile(s)Up	only for option HIST, for HISTO or SHAPE systematic: histogram file name(s) for systematic up variation
HistoFile(s)Down	only for option HIST, for HISTO or SHAPE systematic: histogram file name(s) for systematic down variation
HistoFileSufUp	only for option HIST, for HISTO or SHAPE systematic: suffix of the histogram file names for systematic up variation
HistoFileSufDown	only for option HIST, for HISTO or SHAPE systematic: suffix of the histogram file names for systematic down variation
HistoNameUp	only for option HIST, for HISTO or SHAPE systematic: histogram name for systematic up variation
HistoNameDown	only for option HIST, for HISTO or SHAPE systematic: histogram name for systematic down variation
HistoFolderNameUp	only for option HIST, for HISTO or SHAPE systematic: Name of the folder inside the ROOT file used for the systematic. Example: if your nominal histogram is in "nominal/histo" and the variation is in "sysUp/histo", you can set HistoFolderNameUp to "systUp". Will only replace the first folder in case of nested folders.
HistoFolderNameDown	only for option HIST, for HISTO or SHAPE systematic: Name of the folder inside the ROOT file used for the

Option	Function
	systematic. Example: if your nominal histogram is in "nominal/histo" and the variation is in "sysDown/histo", you can set HistoFolderNameDown to "systDown". Will only replace the first folder in case of nested folders.
HistoFolderSubtractNameUp	only for option HIST, for HISTO or SHAPE systematic: Name of the folder inside the ROOT file used for the systematic to be subtracted e.g. for the JER uncertainty. Example: if your nominal histogram is in "nominal/histo" and the variation is in "sysUp/histo", you can set HistoFolderNameUp to "systUp". Will only replace the first folder in case of nested folders.
HistoFolderSubtractNameDown	only for option HIST, for HISTO or SHAPE systematic: Name of the folder inside the ROOT file used for the systematic to be subtracted e.g. for the JER uncertainty. Example: if your nominal histogram is in "nominal/histo" and the variation is in "sysUp/histo", you can set HistoFolderNameUp to "systUp". Will only replace the first folder in case of nested folders.
HistoNameSufUp	only for option HIST, for HISTO or SHAPE systematic: suffix of the histogram names for systematic up variation
HistoNameSufDown	only for option HIST, for HISTO or SHAPE systematic: suffix of the histogram names for systematic down variation
HistoPathsDownSubtractSample	only for option HIST, paths for the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoPathDownSubtractSample	only for option HIST, path for the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.

Option	Function
HistoPathsUpSubtractSample	only for option HIST, paths for the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoPathUpSubtractSample	only for option HIST, path for the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoFilesUpSubtractSample	only for option HIST, files for the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoFilesDownSubtractSample	only for option HIST, files for the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoFileUpSubtractSample	only for option HIST, file for the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoFileDownSubtractSample	only for option HIST, file for the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoNamesUpSubtractSample	only for option HIST, names for the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoNamesDownSubtractSample	only for option HIST, names for the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoNameUpSubtractSample	only for option HIST, name for the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.

Option	Function
HistoNameDownSubtractSample	only for option HIST, name for the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoNameSufUpSubtractSample	only for option HIST, suffix of the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoNameSufDownSubtractSample	only for option HIST, suffix of the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoFileSufUpSubtractSample	only for option HIST, suffix of the up variation file that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
HistoFileSufDownSubtractSample	only for option HIST, suffix of the down variation file that will be used for subtraction, e.g. needed for full JER scheme with pseudodata.
SampleUp	if set, the syst variation will be built comparing the sample with another sample after all corrections are done; NB: can be used only if the syst affects one sample only
NtuplePath(s)Up	only for option NTUP, for HISTO or SHAPE systematic: ntuple file path(s) for systematic up variation. UNIX-style globbing supported on UNIX-systems (not Windows)
NtuplePath(s)Down	only for option NTUP, for HISTO or SHAPE systematic: ntuple file path(s) for systematic down variation. UNIX-style globbing supported on UNIX-systems (not Windows)
NtuplePathSufUp	only for option NTUP, for HISTO or SHAPE systematic: suffix of the ntuple file paths for systematic up variation

Option	Function
NtuplePathSufDown	only for option NTUP, for HISTO or SHAPE systematic: suffix of the ntuple file paths for systematic down variation
NtupleFile(s)Up	only for option NTUP, for HISTO or SHAPE systematic: ntuple file name(s) for systematic up variation. UNIX-style globbing supported on UNIX-systems (not Windows)
NtupleFile(s)Down	only for option NTUP, for HISTO or SHAPE systematic: ntuple file name(s) for systematic down variation. UNIX-style globbing supported on UNIX-systems (not Windows)
NtupleFileSufUp	only for option NTUP, for HISTO or SHAPE systematic: suffix of the ntuple file names for systematic up variation
NtupleFileSufDown	only for option NTUP, for HISTO or SHAPE systematic: suffix of the ntuple file names for systematic down variation
NtupleName(s)Up	only for option NTUP, for HISTO or SHAPE systematic: ntuple name(s) for systematic up variation
NtupleName(s)Down	only for option NTUP, for HISTO or SHAPE systematic: ntuple name(s) for systematic down variation
NtupleNameSufUp	only for option NTUP, for HISTO or SHAPE systematic: suffix of the ntuple names for systematic up variation
NtupleNameSufDown	only for option NTUP, for HISTO or SHAPE systematic: suffix of the ntuple names for systematic down variation
NtuplePathsUpSubtractSample	only for option NTUP, paths for the up variation that will be used for subtraction, e.g. needed for full JER

Option	Function
	scheme with pseudodata. E.g. the systematic variation can be the MC shifted JER and this option will be used to add the pseudodata variation of JER. Then, the nominal distribution will be added to the difference and considered further in the fit.
NtuplePathsDownSubtractSample	only for option NTUP, paths for the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
NtuplePathUpSubtractSample	only for option NTUP, path for the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
NtuplePathDownSubtractSample	only for option NTUP, path for the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
NtupleFilesUpSubtractSample	only for option NTUP, files for the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
NtupleFilesDownSubtractSample	only for option NTUP, files for the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
NtupleFileUpSubtractSample	only for option NTUP, file for the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
NtupleFileDownSubtractSample	only for option NTUP, file for the down variation that will be used for subtraction, e.g. needed for full JER

Option	Function
	scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
<code>NtupleFileSufUpSubtractSample</code>	only for option NTUP, suffix of the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
<code>NtupleFileSufDownSubtractSample</code>	only for option NTUP, suffix of the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
<code>NtupleNamesUpSubtractSample</code>	only for option NTUP, names for the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
<code>NtupleNamesDownSubtractSample</code>	only for option NTUP, names for the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
<code>NtupleNameUpSubtractSample</code>	only for option NTUP, name for the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
<code>NtupleNameDownSubtractSample</code>	only for option NTUP, name for the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
<code>NtupleNameSufUpSubtractSample</code>	only for option NTUP, suffix of the up variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .

Option	Function
NtupleNameSufDownSubtractSample	only for option NTUP, suffix of the down variation that will be used for subtraction, e.g. needed for full JER scheme with pseudodata. See <code>NtuplePathsUpSubtractSample</code> .
FriendPath(s)Up	only for option NTUP.
FriendPath(s)Down	only for option NTUP.
FriendPathSufUp	only for option NTUP.
FriendPathSufDown	only for option NTUP.
FriendPath(s)UpSubtractSample	only for option NTUP.
FriendPath(s)DownSubtractSample	only for option NTUP.
SampleUp	if set, the syst variation will be built comparing the sample with another sample after all corrections are done; NB: can be used only if the syst affects one sample only
SampleDown	if set, the syst variation will be built comparing the sample with another sample after all corrections are done; NB: can be used only if the syst affects one sample only
WeightUp	only for option NTUP, for HISTO or SHAPE systematic: weight for systematic up variation; it will be multiplied with the <code>MCweight</code> defined in the Job and Region blocks, however it will not be multiplied with the <code>MCweight</code> defined in the sample block of the reference sample (for this, use <code>WeightSufUp</code> instead)
WeightDown	only for option NTUP, for HISTO or SHAPE systematic: weight for systematic down variation; it will be multiplied with the <code>MCweight</code> defined in the Job and

Option	Function
	Region blocks, however it will not be multiplied with the <code>MCweight</code> defined in the sample block of the reference sample (for this, use <code>WeightSufDown</code> instead)
WeightSufUp	only for option NTUP, for HISTO or SHAPE systematic: additional weight for systematic up variation (also multiplied with the <code>MCweight</code> acting on the nominal sample)
WeightSufDown	only for option NTUP, for HISTO or SHAPE systematic: additional weight for systematic down variation (also multiplied with the <code>MCweight</code> acting on the nominal sample)
IgnoreWeight	if set to <code>TRUE</code>, weights from Job, Region and Sample will be ignored. If set to a string, that particular substring will be ignored from the weight string
Symmetrisation	can be ONESIDED, TWOSIDED, ABSMEAN and MAXIMUM (...); for no symmetrisation, skip the line; ONESIDED = only one variation provided (e.g. up variation), down variation will be added as mirrored version, TWOSIDED = (up-down)/2 variation is calculated bin-by-bin, this is used as up variation and then mirrored to down variation, ABSMEAN = $((\text{abs}(\text{up}) + \text{abs}(\text{down})) / 2)$ can be used when both variations have the same sign, MAXIMUM = take variations with larger abs value w.r.t nominal and take this in each bin, this might flip signs!, MAXIMUMKEEPSIGN = looks at the up variation, if it is larger than down, no change is made, if down variation is larger, the up variation is modified to be nom - down, the down variation is then mirrored (this preserves the sign of the up variation, just modifies the size). PLEASE BE VERY CAREFUL WHEN USING THESE SYMMETRISATION OPTIONS AS THEY MIGHT MODIFY THE HISTOGRAM IN AN UNEXPECTED WAY. ALWAYS CHECK THE PLOTS IN THE SYSTEMATICS FOLDER

Option	Function
Smoothing	smoothing code to apply; historically 40 is was used for the MAXVARIATION smoothing (maximum 4 slope changes); for TTBARRESONANCE 40 means uncorrelated bins, 400 means correlated; for no smoothing, skip the line
OverallUp	for OVERALL systematic: the relative "up" shift (0.1 means +10%)
OverallDown	for OVERALL systematic: the relative "down" shift (-0.1 means -10%)
ScaleUp	for OVERALL, HISTO or SHAPE systematic: scale difference between "up" and nominal by a factor, or different factors for different regions (with the syntax <code>region1:1.2,region2:0.9</code>)
ScaleDown	for OVERALL, HISTO or SHAPE systematic: scale difference between "down" and nominal by a factor, or different factors for different regions (with the syntax <code>region1:1.2,region2:0.9</code>)
ReferenceSample	if this is specified, the syst variation is evaluated w.r.t. this reference sample (often a GHOST sample) instead of the nominal, and then the relative difference is propagated to nominal.
ReferenceSmoothing	if this is specified, the syst variation is smoothed wrt a specified sample (Must appear in <code>Samples</code> for this syst) and then the smoothed variations are copied bin by bin to all other samples specified. Useful when <code>Morphing</code> is used
ReferencePruning	if this is specified, the syst variation is pruned wrt a specified sample (Must appear in <code>Samples</code> for this syst) and then the same pruning is applied to all specified samples

Option	Function
DropShapeIn	Specify regions or samples where you want the smoothing / pruning to be forced to drop the shape and keep only norm. When <code>all</code> is used, the shape is dropped for all regions and all samples. DANGER: only use this if you are very sure this is what you need, it usually is not!
DropNorm	Same as the previous one, but to drop the norm and keep only the shape. DANGER: only use this if you are very sure this is what you need, it usually is not! This is not equivalent to re-normalizing systematic variations in the presence of a normalization factor as this drop acceptance effects between regions.
DropNormSpecial	Only effective when <code>CombineName</code> is used to combine special systematics. A list of region names can be specified to drop the normalisation for the combined systematic in those regions. This option has to be used for at least one of the systematics that are being combined. DANGER: only use this if you are very sure this is what you need, it usually is not! This is not equivalent to re-normalizing systematic variations in the presence of a normalization factor as this drop acceptance effects between regions.
KeepNormForSamples	list of samples (or sum of samples, in the form <code>smp1+smp2</code>), comma separated, for which the systematic gets shape only in each region
DummyForSamples	list of samples, comma separated, for which the systematic gets created as a dummy one (syst variation = nominal); useful when used in combination with <code>KeepNormForSamples</code>
NoPruning	Force the code to skip pruning for the systematic

Option	Function
PreSmoothing	if set to TRUE, a TH1::Smooth-based smoothing is applied, prior to the usual smoothing (if set)
SmoothingOption	if not set smoothing option from Job option is taken. Choose which smoothing option to use for this systematic (this will overwrite the general smoothing option for this), allowed parameters are: MAXVARIATION (default), TTBARRESONANCE (see also FAQ section), COMMONTOOLSMOOTHMONOTONIC, COMMONTOOLSMOOTHPARABOLIC, KERNELRATIOUNIFORM, KERNELDELTA GAUSS or KERNELRATIOGAUSS.
Decorrelate	decorrelate systematic, can take values REGION (decorrelate across regions), SAMPLE (decorrelate across samples), SHAPEACC (decorrelate shape and acceptance effects); can be used to change behaviour of a systematic without having to re-run the n or b step
ForceShape	Can be: NOSHAPE (default) - no changes to the uncertainty, LINEAR will create a linear shape from left up variation to bottom down variation ased on the original up and down variations or TRIANGULAR that will create a triangular shape that is increasing until the middle of the distribution and then decreasing back to 0. The other variation is symmetrised.

Morphing block settings

Option	Function
FitFunction	Tells the code what fit function is used in each bin. Can be either LINEAR , QUADRATIC , CUBIC , QUADRATIC , COSINE , or a string that can be interpreted in TFormula.

Option	Function
ParamStartValue	Allows to "help" the fit to converge by setting the starting values of the parameters. Syntax: <code>target@regionName@binIndex(from 0):parameterIndex:value</code> , comma separated (example: <code>signalSample@Region1@0:0:1.4,Region1@0:1:2,Region2@0:1:2</code>). The following shorthands are available: <code>ALL</code> for signal targets, <code>SIGNAL / CONTROL / VALIDATION</code> for regions (can also be combined like <code>SIGNALCONTROL</code>), <code>ALL</code> for bins.
ParamLimits	Allows to "help" the fit to converge by setting the limits of the parameters. Syntax: <code>target@regionName@binIndex(from 0):parameterIndex:min:max</code> , comma separated (example: <code>signalSample@Region1@0:0:0:2,Region1@0:1:0:2,Region2@0:1:0:2</code>). The following shorthands are available: <code>ALL</code> for signal targets, <code>SIGNAL / CONTROL / VALIDATION</code> for regions (can also be combined like <code>SIGNALCONTROL</code>), <code>ALL</code> for bins.
ParamRangeScale	Sets the limits of the parameters used in the formulae for the TRExFitter fit. The range is set to <code>value</code> times max template - min template. <code>value</code> can be set with this option, default is 0.5
Chi2WarningLimit	Sets the limit for printing warnings from the individual bin fits if χ^2/NDF is larger than this value. Default is 1.5
Make1DProjections	If set to <code>TRUE</code> , 1D projections of 2D parametrisation are plotted and compared to template yields in subfolder <code>Templates</code> . Default is <code>FALSE</code> .
Expressions	a way to to replace the morphing parameters with another formula; analogous to the NormFactor option Expression, but accepts a list of expressions, with this format <code><param1>=<expression>:<dependencies></code> , <code><param22>=<expression>:<dependencies></code> [example: <code>"topWidth"="0.5*topMass": "topMass[150,200]</code>] (NB: mandatory usage of quotation marks in case of expressions with more that one argument, as in the example). Can reparametrize both of the parameters. The replaced expressions will appear in the <code>Templates/Parametrization.txt</code> file - check it carefully.

EFTConfig block settings

Option	Function
EFTOrder	can be ALL (default) or LIN to fit only the linear-terms on the EFT. The parameterisation will always calculate all terms, but for LIN the Nuisance Parameters will only consider the linear terms in the fit.
SplitSamplePerBin	if set to <code>TRUE</code> (default is <code>FALSE</code>) will proceed with the EFT fit where a new sample for each bin will be created with an attached NormFactor. Setting to <code>FALSE</code> will run an EFT fit where ShapeFactors are attached to the SM reference samples and these are then reparametrised based on the EFT parameterisation. Setting <code>TRUE</code> is supported ONLY FOR NTUPLE PROCESSING
ParameterMinMax	Set the minimum and maximum value used for the EFT parameters in the workspace. Syntax: <code>parameter1:minimum:maximum,parameter2:minimum:maximum</code> . The default minimum and maximum values are: -50,50.
ExcludedParameterRegionSample	Allows to remove the EFT parameter(s) from a given Region and Sample. Syntax: <code>parameter1:region1:sample1,parameter2:region2:sample2</code> . Supports wildcards (<code>*</code>) for the region and sample setting.
PlotRatio	If set to <code>TRUE</code> (default), will produce a ratio plot in the bottom panel for the EFT plots.

Multi-fit

These options are for multi-fits, performed with action `m`.

Multi-fit Job block settings

Option	Function
Label	the label which will be shown on plots
DebugLevel	0 = prints only Warning and Errors, 1 = additionally prints Info messages, 2 = additionally prints Debug messages, >2 additionally prints Verbose messages. For option <2 RooFit/Roostats messages will be heavily suppressed
LoggingFormat	Can be set to 'NONE' to have no logging header. Otherwise a list of options from: 'TIME','DATE', 'FILELINES' and 'FUNCTION' is allowed. Each option turns on the logging header for each line presenting, in order, the time of the day, the day, name of the file and lines where the message is printed from, the function name where the message is printed from. Default is 'TIME,DATE,FILELINES'. Please, make sure there are no white spaces between the commas as the info is pre-read.
OutputDir	the name of the output directory
LumiLabel	the luminosity label that will be shown on the plots
CmeLabel	the center of mass energy label that will be shown on the plots
CombiLabel	the name of the combination that will be shown on the plots, default is "Combined"
SaveSuf	added to file name of histograms, for usage with hupdate (equivalent to command line option)
ShowObserved	can be TRUE or FALSE, flag to turn on/off the observed values on the plots
LimitTitle	the title for limit that will be shown on the plots

Option	Function
POITitle	the title of the POIs, separated by commas, that will be shown on X axis
CompareLimits	can be TRUE or FALSE, flag to compare to Limit values
ComparePOI	can be TRUE or FALSE, flag to compare to POI values
ComparePulls	can be TRUE or FALSE, flag to compare to pulls values
CompareEFT	if set to <code>TRUE</code> (default), will produce EFT comparison for the EFT fits
PlotCombCorrMatrix	can be set to TRUE or FALSE, flag to build correlation matrix from the combined systematics
CorrelationThreshold	Threshold used to draw the correlation matrix (only systematics with at least one correlation larger than than draw) (0.05:5%)
Combine	can be TRUE or FALSE, set to TRUE if you want to perform actual combination (followed by <code>mwf</code>)
Compare	can be TRUE or FALSE, set to TRUE if you want to compare values
StatOnly	can be TRUE or FALSE, set to TRUE if the fits are stat only fits
IncludeStatOnly	can be TRUE or FALSE, set to TRUE if you want to include stat only fits
POIName	the name of the POI in the configs

Option	Function
POIRange	the range of the chosen POIs, in the form of <code>min1:max1,min2:max2</code> , where the number of options needs to match the number of POIs
LimitMax	set maximum value for the limit
POIPrecision	comma separated strings, set precision of the POIs
DataName	can be "obsData", "asimovData", or custom string, if nothing is specified the observed data will be used
FitType	can be SPLUSB or BONLY
NPCategories	comma separated list of NP categories
SetRandomInitialNPval	provide a float
SetRandomInitialNPvalSeed	provide an int
NumCPU	a number of CPU cores used for the fit
FastFit	can be TRUE or FALSE
FastFitForRanking	can be TRUE or FALSE
NuisParListFile	Name of file containing list of nuisance parameters, with one parameter per line, and names just like in the <code>Fits/*.txt</code> file. The order will be used for the plots created with <code>ComparePulls</code> .
PlotSoverB	if set to TRUE will plot signal over background plots
SignalTitle	a title of the signal for the plots
FitResultsRootFile	a name of the ROOT file with fit results

Option	Function
LimitsFile	a name of the file with limits results
BonlySuffix	a suffix of the background only fits
ShowSystForPOI	can be TRUE or FALSE, set to TRUE if you want to show systematics for POI
doLHscan	comma separated list of NP(or POIs) to run LH scan, if first parameter is "all" it will be run for all NP
do2DLHscan	produces 2D likelihood scan between the chosen parameters. Syntax: "paramX1,paramY1:paramX2,paramY2". Warning takes long time. You can reduce the number of steps via <code>LHscanSteps</code> . Alternatively you can split up the 2D scan in slices with <code>LHscanStep</code> and <code>LHscanStepY</code> command line options
LHscanMin	minimum value for the LH scan on x-axis (default is Norm min). This also effect the x-axis in a 2D scan
LHscanMax	maximum value for the LH scan on x-axis (default is Norm max). This also effect the x-axis in a 2D scan
LHscanSteps	number of steps on the LH scan (default is 30) . This also effect the x-axis in a 2D scan
LHscanMinY	minimum value for the 2D-LH scan on y-axis (default it Norm min)
LHscanMaxY	maximum value for the 2D-LH scan on y-axis (default is Norm max)
LHscanStepsY	number of steps on the LH scan in y-direction (default is 30, but if not specified it uses LHscanSteps)

Option	Function
PlotOptions	same as for "standard" fits
POILabel	names of the POIs shown on plots (size and order has to match the POIs defined), default is <code>#\mu</code>
POINominal	values of the nominal (SM) prediction for POIs (size and order has to match the POIs defined), default is <code>1</code>
ShowTotalOnly	If set to TRUE will show only total uncertainty on the POI plots. Default is FALSE
POIAsimov	Sets the the values of the POIs for the fit (size and order has to match the POIs defined). Needs to be used with <code>DataName: asimovData</code> . Default is 1.
POIInitial	Sets the initial value of the POIs for the fit. Comma separated list of values. The size and order has to match the definition of POIs.
HEPDataFormat	If set to <code>TRUE</code> will produce output in HEPData yaml format.
UheppFormat	If set to <code>TRUE</code> will produce plots in Uhepp format.
FitStrategy	Set Minuit2 fitting strategy, can be: 0, 1, 2 or 3. If negative value is set the default is used (1)
BinnedLikelihoodOptimization	Can be set to TRUE or FALSE (default). If se to TRUE, will use the <code>BinnedLikelihood</code> optimisation of RooFit that has significant speed improvements, but results in less stable correlation matrix computation
UsePOISinRanking	If set to <code>TRUE</code> (default is <code>FALSE</code>) will include POIs as NPs in the ranking

Option	Function
ShiftGlobalObservablesInRanking	If set to <code>TRUE</code> (default is <code>FALSE</code>) will run ranking by shifting the global observables instead of shifting and fixing a nuisance parameter
UseHesseBeforeMigrad	If set to <code>TRUE</code> (default is <code>FALSE</code>) will run hesse() method before migrad, this can help with convergence in some cases
Regions	A comma separated list of regions to be considered. If not provided, all regions are used. Does NOT impact the workspace combination (fit).
MultiStageFit	Same setting as in the standalone fit
NPCutOff	Same setting as in the standalone fit
PlotUnfoldedData	Only valid when <code>FitType: UNFOLDING</code> is used. Will produce the unfolded data plot using the MC truth style/inputs from the first config file used.
UnfoldingShowStatOnlyError	If set to <code>TRUE</code> (default is <code>FALSE</code>), will add stat-only error band on the unfolded plots. <i>Requires to run IncludeStatOnly: TRUE</i> to get the stat only results. When Using <code>IncludeStatOnly: TRUE</code> , this option defaults to <code>TRUE</code> .
UnfoldingShowUncertaintyBreakdown	If set to <code>TRUE</code> (default is <code>FALSE</code>), when running an UNFOLDING fit, this will produce a histogram visualising the grouped impact. Is produced during <code>GroupedImpact=combine</code> step.
UnfoldingUncertaintyBreakdownTotal	If set to <code>TRUE</code> (default is <code>TRUE</code>), will show the total uncertainty when <code>UnfoldingShowUncertaintyBreakdown</code> is set. When set to <code>FALSE</code> will only show total systematics instead of total (stat+syst) uncertainty.

Option	Function
PlotGammas	If set to <code>TRUE</code> (default is <code>FALSE</code>), will produce the gamma plot for the combined fit during the 'mf' step.
BlindedParameters	Blinded parameters. The ones in the individual config files are automatically propagated
RankingPOIName	Custom name for the POI for ranking plots. Comma separated list. The number and order must match the POI setting. Default is <code>#mu</code>
RankingPOIAxisScale	Order of magnitude of the scale for the POI axis for ranking plots. Comma separated list. The number and order must match the POI setting. Default is 1.0
RankingUpperAxisNdivision	Standard ROOT Ndivision option for upper axis for ranking plots. Comma separated list. The number and order must match the POI setting. Default is 510
MaximumNumberFCNcalls	Maximum number of function calls during minimisation. If set to negative value (default) 500 x N will be used for N < 100 (RooFit default) and 5 x N ² for N > 100, where N is the number of the free parameters in the fit.
ToleranceScale	Value to be used to scale the default tolerance (set to 0.01) for the fit minimisation. The default is 1, i.e. the default tolerance is used.
UseAutoDiff	If set to <code>TRUE</code> (default is <code>FALSE</code>) the likelihood minimisation will use <code>clad</code> to calculate the derivatives analytically for the gradient. All HistFactory models are supported, also some extensions e.g. using simple polynomials for reparametrisation (such as EFT fit)
NLLOffset	Set the likelihood offsetting mode. Allowed values are: <code>INITIAL</code> (default), which offsets the likelihood

Option	Function
	by its initial value, <code>BIN</code> , which applies a bin-by-bin offset with a template histogram model based on the observed data, or <code>NONE</code> . Refer to <code>RooFit::RooAbsPdf::createNLL()</code> documentation for more information.
FitToys	if set to $N > 0$, N toys are generated and fitted
ToysHistoNbins	If FitToys is used, set number of bins for toys histogram output
ToysSeed	Set initial seed for the toys generation. Useful for generation of multiple independent parallel toys jobs, the outputs of which should be combined later by the user. Default is 1234. NOTE: Parallel jobs will each use a range of seeds from (ToySeed)-> (ToySeed+NToys). Each job must therefore use a unique seed starting beyond the range of the previous job - otherwise the toys will statistically overlap.
FitToysRandomInitialNormFactorValues	Allows to set random initial values for the NormFactors during the toys fitting (in the <code>f</code> step). This might be needed for fits with multiple minima, e.g. EFT fit. Syntax: comma separated list of <code>normFactorName:minimum:maximum</code> , where <code>normFactorName</code> is the name of the NF to be randomised, <code>minimum</code> and <code>maximum</code> are the values to be used in the random uniform generation for the starting points. E.g. <code>ctg:-1:1,ctW:-0.5:0.5</code>

Option	Function
FitToysLHscanCondition	Allows to run LH scan for a toy dataset falling into some range of the fitted values during the toys fitting (in the <code>f</code> step). Syntax: comma separated list of <code>normFactorName:minimum:maximum</code> , where <code>normFactorName</code> is the name of the NF to be randomised, <code>minimum</code> and <code>maximum</code> are the values where the given toys should fall. Will run the LH scan only on the first "hit". E.g. <code>ctg:-1:1,ctW:-0.5:0.5</code>
ToysLHScanForAll	If set to <code>TRUE</code> (default is <code>FALSE</code>), will run LHscan for all POIs for all toys (this can be very slow!). No plots will be produced, but the results will be stored in a ROOT file in the <code>LHoodPlots/</code> folder
ToysOnlyStatFluctuation	If set to <code>TRUE</code> (default is <code>FALSE</code>), when generating the toys, only the statistical fluctuations will be considered and not the systematic uncertainties. The fit for each toy will use the full model. To run StatOnly toys with StatOnly fits, generate StatOnly workspace and run the toys

Multi-fit `Fit` block settings

Option	Function
Options	additional options, accepting only float as arguments - useful for adding your functionalities & flags in a quick way, since they need minimal changes in the code) ...
Label	the label of the values from this config that will be shown on the plots
LoadSuf	
ConfigFile	the path to the config file that you want to combine/compare

Option	Function
Workspace	the path to the workspace that you want to combine/compare
FitResultsRootFile	the path to the file with fit results
LimitsFile	the path to the file with limits results
POIName	the names of the POIs
Directory	the path to the directory
InputName	the name of the input
UseInFit	Set to <code>TRUE</code> by default, if set to <code>FALSE</code> will use the config in comparison plots, but will not be used to create the workspace and will not be used in the fit
UseInComparison	Set to <code>TRUE</code> by default, if set to <code>FALSE</code> will not use the config in comparison plots.

Multi-fit `Limit` block settings

Option	Function
LimitType	can be ASYMPTOTIC (default) or TOYS (TOYS uses WS as in the 'f' step)
LimitBlind	can be TRUE or FALSE (TRUE means that ALL regions are blinded)
SignalInjection	if set to TRUE, expected signal with signal injection is evaluated
SignalInjectionValue	Value for the injected signal

Option	Function
POI	Name of the POI to be used in case of multiple POIs
ParamName	Name for the parameter in the output ROOT file
ParamValue	Value of the parameter in the output file (e.g. 172.5 for top mass)
OutputPrefixName	Prefix for the output ROOT file
ConfidenceLevel	Confidence level for the CLs. Default is 0.95
SplusBToys	Only for limit type <code>TOYS</code> . Set the number of toys for SplusB
BonlyToys	Only for limit type <code>TOYS</code> . Set the number of toys for Bonly
ScanSteps	Only for limit type <code>TOYS</code> . Number of steps for limit scanning
ScanMin	Only for limit type <code>TOYS</code> . Min for limit scanning
ScanMax	Only for limit type <code>TOYS</code> . Max for limit scanning
LimitPlot	Only for limit type <code>TOYS</code> . If set to <code>TRUE</code> (default) will produce brazilian-style plot in the Limits/ folder
LimitFile	Only for limit type <code>TOYS</code> . If set to <code>TRUE</code> (default) will produce ROOT file in the Limits/ folder with information per point
FitStrategy	Set Minuit2 Hesse strategy for limit extraction, can be: 0, 1, 2 or 3. The default is 1 (No <code>Hesse</code>). Accuracy and computational cost increase with this value.
ToysSeed	Seed for the random numbers in case of <code>TOYS</code> type
UseNLLwithoutOffsetInLHscan	If set to <code>TRUE</code> (default) will use NLL values that dont use offset subtraction

Option	Function
UseHesse	Can be set to <code>TRUE</code> (default) or <code>FALSE</code> . Tells the code to use <code>hesse()</code> call in the minimisation during the <code>f</code> step
ToysUsexRooFit	Only for limit type <code>TOYS</code> . Use the xRooFit backend to compute the toy-based limits. Valid option are <code>FALSE</code> (default), <code>TRUE</code> , and <code>AUTO</code> . A value of <code>TRUE</code> uses the specified <code>SplusBToys</code> , <code>BonlyToys</code> , <code>ScanSteps</code> , <code>ScanMin</code> and <code>ScanMax</code> values to configure the scan, while a value of <code>AUTO</code> uses xRooFit's auto-configuration mode.
UseAutoDiff	If set to <code>TRUE</code> (default is <code>FALSE</code>) the likelihood minimisation will use <code>clad</code> to calculate the derivatives analytically for the gradient. All HistFactory models are supported, also some extensions e.g. using simple polynominals for reparametrisation (such as EFT fit)

Multi-fit `Significance` block settings

Option	Function
SignificanceType	can be <code>ASYMPTOTIC</code> (default) or <code>TOYS</code> (<code>TOYS</code> uses <code>WS</code> as in the 'f' step)
SignificanceBlind	can be <code>TRUE</code> or <code>FALSE</code> (<code>TRUE</code> means that ALL regions are blinded)
POI	Name of the POI to be used in case of multiple POIs
POIAsimov	value of the POI to inject in the Asimov dataset in <code>LimitBlind</code> is set to <code>TRUE</code>
ParamName	Name for the parameter in the output ROOT file
ParamValue	Value of the parameter in the output file (e.g. 172.5 for top mass)
OutputPrefixName	Prefix for the output ROOT file

Option	Function
SplusBToys	Only for significance type <code>TOYS</code> . Set the number of toys for SplusB. Default value is 0 as this is not needed for significance, it is only useful for plots for comparison
BonlyToys	Only for significance type <code>TOYS</code> . Set the number of toys for Bonly
ToysSeed	Seed for the random numbers in case of <code>TOYS</code> type
SignificancePlot	Only for significance type <code>TOYS</code>
ToysUsexRooFit	Only for limit type <code>TOYS</code> . Use the xRooFit backend to compute the toy-based limits. Valid option are <code>FALSE</code> (default) or <code>TRUE</code>
FitStrategy	Set Minuit2 fitting strategy, can be: 0, 1, 2 or 3. If negative value is set the default is used (1)
UseAutoDiff	If set to <code>TRUE</code> (default is <code>FALSE</code>) the likelihood minimisation will use <code>clad</code> to calculate the derivatives analytically for the gradient. All HistFactory models are supported, also some extensions e.g. using simple polynominals for reparametrisation (such as EFT fit)

Unfolding fit

These options are for unfolding fits, set up with action `u` .

UnfoldingSample block

Option	Function
Title	for plots
FillColor	for plots
FillColorRGB	for plots

Option	Function
LineColor	for plots
LineColorRGB	for plots
UnfoldingName	In case of multiple Unfolding blocks, specify which unfolding this sample belongs to
ResponseMatrixFile(s)	same as for Job but per sample
ResponseMatrixFileSuff(s)	same as for Job but per sample
ResponseMatrixName(s)	same as for Job but per sample
ResponseMatrixNameSuff(s)	same as for Job but per sample
ResponseMatrixPath(s)	same as for Job but per sample
ResponseMatrixPathSuff(s)	same as for Job but per sample
AcceptanceFile(s)	same as for Job but per sample
AcceptanceFileSuff(s)	same as for Job but per sample
AcceptanceName(s)	same as for Job but per sample
AcceptanceNameSuff(s)	same as for Job but per sample
AcceptancePath(s)	same as for Job but per sample
AcceptancePathSuff(s)	same as for Job but per sample
SelectionEffFile(s)	same as for Job but per sample
SelectionEffFileSuff(s)	same as for Job but per sample

Option	Function
SelectionEffName(s)	same as for Job but per sample
SelectionEffNameSuff(s)	same as for Job but per sample
SelectionEffPath(s)	same as for Job but per sample
SelectionEffPathSuff(s)	same as for Job but per sample
MigrationFile(s)	same as for Job but per sample
MigrationFileSuff(s)	same as for Job but per sample
MigrationName(s)	same as for Job but per sample
MigrationNameSuff(s)	same as for Job but per sample
MigrationPath(s)	same as for Job but per sample
MigrationPathSuff(s)	same as for Job but per sample
Regions	Comma separated list of regions
Type	Can be <code>STANDARD</code> (default) or <code>GHOST</code> . <code>GHOST</code> should be used only for samples that will then be used as <code>ReferenceSample</code>
Gammas	Can be <code>SEPARATED</code> (default) or <code>DISABLED</code> . <code>SEPARATED</code> will use <code>SeparateGammas</code> for each subsample of signal - effectively adding one NP per response matrix bin. <code>DISABLED</code> will remove gammas from the signal completely
SubSampleRegions	used to allow samples generated from certain truth bins to populate only a set of regions; example usage: <code>1:"region1",1:"region2",2:"region3"</code> (here truth bin 1 will be folded to regions 1 and 2 only, while truth bin 2 will populate only region 3)

Option	Function
NoPruning	propagated to all subsamples, same as for Sample

UnfoldingSystematic block

Option	Function
Samples	UnfoldingSamples affected by this uncertainty
Regions	Regions affected by this uncertainty
NuisanceParameter	Used to correlate systematics
Type	HISTO/OVERALL/SHAPE/STAT
Title	for plots
Category	same as for Systematic
SubCategory	same as for Systematic
Symmetrisation	same as for Systematic
SmoothingOption	same as for Systematic
Smoothing	same as for Systematic
UnfoldingName	In case of multiple Unfolding blocks, specify which unfolding this systematic belongs to
ResponseMatrixFile(s)Up	self-explanatory
ResponseMatrixFileSuff(s)Up	self-explanatory

Option	Function
ResponseMatrixName(s)Up	self-explanatory
ResponseMatrixFolderNameUp	self-explanatory
ResponseMatrixPath(s)Up	self-explanatory
ResponseMatrixPathSuff(s)Up	self-explanatory
ResponseMatrixNameSuff(s)Up	self-explanatory
ResponseMatrixFile(s)Down	self-explanatory
ResponseMatrixFileSuff(s)Down	self-explanatory
ResponseMatrixName(s)Down	self-explanatory
ResponseMatrixNameSuff(s)Down	self-explanatory
ResponseMatrixFolderNameDown	self-explanatory
ResponseMatrixPath(s)Down	self-explanatory
ResponseMatrixPathSuff(s)Down	self-explanatory
AcceptanceFile(s)Up	self-explanatory
AcceptanceFileSuff(s)Up	self-explanatory
AcceptanceName(s)Up	self-explanatory
AcceptanceNameSuff(s)Up	self-explanatory
AcceptanceFolderNameUp	self-explanatory
AcceptancePath(s)Up	self-explanatory

Option	Function
AcceptancePathSuff(s)Up	self-explanatory
AcceptanceFile(s)Down	self-explanatory
AcceptanceFileSuff(s)Down	self-explanatory
AcceptanceName(s)Down	self-explanatory
AcceptanceNameSuff(s)Down	self-explanatory
AcceptanceFolderNameDown	self-explanatory
AcceptancePath(s)Down	self-explanatory
AcceptancePathSuff(s)Down	self-explanatory
SelectionEffFile(s)Up	self-explanatory
SelectionEffFileSuff(s)Up	self-explanatory
SelectionEffName(s)Up	self-explanatory
SelectionEffNameSuff(s)Up	self-explanatory
SelectionEffFolderNameUp	self-explanatory
SelectionEffPath(s)Up	self-explanatory
SelectionEffPathSuff(s)Up	self-explanatory
SelectionEffFile(s)Down	self-explanatory
SelectionEffFileSuff(s)Down	self-explanatory
SelectionEffName(s)Down	self-explanatory

Option	Function
SelectionEffNameSuff(s)Down	self-explanatory
SelectionEffFolderNameDown	self-explanatory
SelectionEffPath(s)Down	self-explanatory
SelectionEffPathSuff(s)Down	self-explanatory
MigrationFile(s)Up	self-explanatory
MigrationFileSuff(s)Up	self-explanatory
MigrationName(s)Up	self-explanatory
MigrationNameSuff(s)Up	self-explanatory
MigrationFolderNameUp	self-explanatory
MigrationPath(s)Up	self-explanatory
MigrationPathSuff(s)Up	self-explanatory
MigrationFile(s)Down	self-explanatory
MigrationFileSuff(s)Down	self-explanatory
MigrationName(s)Down	self-explanatory
MigrationNameSuff(s)Down	self-explanatory
MigrationFolderNameDown	self-explanatory
MigrationPath(s)Down	self-explanatory
MigrationPathSuff(s)Down	self-explanatory

Option	Function
ResponseMatrixPath(s)UpSubtractSample	For special systematics, e.g. full JER
ResponseMatrixPath(s)DownSubtractSample	For special systematics, e.g. full JER
ResponseMatrixFile(s)UpSubtractSample	For special systematics, e.g. full JER
ResponseMatrixFile(s)DownSubtractSample	For special systematics, e.g. full JER
ResponseMatrixName(s)UpSubtractSample	For special systematics, e.g. full JER
ResponseMatrixName(s)DownSubtractSample	For special systematics, e.g. full JER
ResponseMatrixFolderNameUpSubtractSample	For special systematics, e.g. full JER
ResponseMatrixFolderNameDownSubtractSample	For special systematics, e.g. full JER
MigrationPath(s)UpSubtractSample	For special systematics, e.g. full JER
MigrationPath(s)DownSubtractSample	For special systematics, e.g. full JER
MigrationFile(s)UpSubtractSample	For special systematics, e.g. full JER
MigrationFile(s)DownSubtractSample	For special systematics, e.g. full JER
MigrationName(s)UpSubtractSample	For special systematics, e.g. full JER
MigrationName(s)DownSubtractSample	For special systematics, e.g. full JER
MigrationFolderNameUpSubtractSample	For special systematics, e.g. full JER
MigrationFolderNameDownSubtractSample	For special systematics, e.g. full JER
AcceptancePath(s)UpSubtractSample	For special systematics, e.g. full JER
AcceptancePath(s)DownSubtractSample	For special systematics, e.g. full JER

Option	Function
AcceptanceFile(s)UpSubtractSample	For special systematics, e.g. full JER
AcceptanceFile(s)DownSubtractSample	For special systematics, e.g. full JER
AcceptanceName(s)UpSubtractSample	For special systematics, e.g. full JER
AcceptanceName(s)DownSubtractSample	For special systematics, e.g. full JER
AcceptanceFolderNameUpSubtractSample	For special systematics, e.g. full JER
AcceptanceFolderNameDownSubtractSample	For special systematics, e.g. full JER
SelectionEffPath(s)UpSubtractSample	For special systematics, e.g. full JER
SelectionEffPath(s)DownSubtractSample	For special systematics, e.g. full JER
SelectionEffFile(s)UpSubtractSample	For special systematics, e.g. full JER
SelectionEffFile(s)DownSubtractSample	For special systematics, e.g. full JER
SelectionEffName(s)UpSubtractSample	For special systematics, e.g. full JER
SelectionEffName(s)DownSubtractSample	For special systematics, e.g. full JER
SelectionEffFolderNameUpSubtractSample	For special systematics, e.g. full JER
SelectionEffFolderNameDownSubtractSample	For special systematics, e.g. full JER
ReferenceSample	If a name of one of the UnfoldingSamples is provided, the response matrix for this systematic will be modified by subtracting difference between the reference UnfoldingSample and the nominal one

Option	Function
OverallUp	for OVERALL type only: the relative "up" shift (0.1 means +10%)
OverallDown	for OVERALL type only: the relative "down" shift (-0.1 means -10%)
CombineName	same as the Systematic version
CombineType	same as the Systematic version
DropShapeIn	Comma separated list of regions where the shape component from all subsamples is dropped (the effect will be a pure normalisation one) for a given systematic
DropNorm	Comma separated list of regions where the normalisation component from all subsamples is dropped (the effect will be a pure shape one) for a given systematic. DANGER: only use this if you are very sure this is what you need, it usually is not! This is not equivalent to re-normalizing systematic variations in the presence of a normalization factor as this drop acceptance effects between regions.
ScaleUp	for OVERALL, HISTO or SHAPE systematic: scale difference between "up" and nominal by a factor, or different factors for different regions (with the syntax <code>region1:1.2, region2:0.9</code>)
ScaleDown	for OVERALL, HISTO or SHAPE systematic: scale difference between "down" and nominal by a factor, or different factors for different regions (with the syntax <code>region1:1.2, region2:0.9</code>)

Unfolding block

Option	Function
MatrixOrientation	Can be TRUTHONHORIZONTAL/TRUTHONVERTICAL, self explanatory
TruthDistributionPath	Globally set the path, will be overwritten by TruthSample
TruthDistributionFile	Globally set the path, will be overwritten by TruthSample
TruthDistributionName	Globally set the path, will be overwritten by TruthSample
NumberOfTruthBins	Number of truth bins (can be only one truth distribution but can have more reco distributions - regions)
Tau	Tikhonov regularization parameter, use as e.g. <code>1:2,3:1.7</code> , this will set the parameter to bins 1 and 3, with values 2 and 1.7 respectively; alternatively, a single value can be given, which will be used for all the unfolding norm-factors
TitleX	Name of the title on unfolded plots
TitleY	Name of the title on unfolded plots
RatioYmax	Used for SetRangeUser
RatioYmin	Used for SetRangeUser
LogX	Set to TRUE to use logarithmic scale for unfolded plots
LogY	Set to TRUE to use logarithmic scale for unfolded plots
TitleOffsetX	scale of the original offset (default is 1)
TitleOffsetY	scale of the original offset (default is 1)

Option	Function
ErrorBandColor	colour of the total error band in the unfolded plots, default is kGray+1
StatErrorBandColor	colour of the stat-only error band in the unfolded plots, default is kGray
RatioLineStyle	style of the line in the ratio plot at Y=1, default is 2
RatioLineColor	colour of the line in the ratio plot at Y=1, default is kRed
RatioLineWidth	width of the line in the ratio plot at Y=1, default is 2. If set to 0 the line will not be added to the plot
LegendXposition	X position of the legend in the unfolded plot, default is 0.5
LegendTextSize	text size of the legend, default uses trex style default
MinRangeY	Minimum value for the Y axis range for the unfolding plot. Default value is 0.0001
ScaleRangeY	scale of the Y axis for the upper panel of the result plot (will use this value times the maximum). If set to negative value, wil use default (<code>1.5</code> for linear scale, <code>1e6</code> for log scale)
UnfoldingResultMin	Minimum for the normalisation factors for bins (Default is 0)
UnfoldingResultMax	Maximum for the normalisation factors for bins (Default is 2)
NominalTruthSample	Name of the TruthSample that should be used as nominal for folding. Has to be set!
MigrationTitleX	name of the title on migration/response matrix plots - X axis. If a single name is provided, this will be used in all regions. Can also provide per region title with <code>region1:name1,region2:name2</code> syntax

Option	Function
MigrationTitleY	name of the title on migration/response matrix plots - Y axis. If a single name is provided, this will be used in all regions. Can also provide per region title with <code>region1:name1, region2:name2</code> syntax
MigrationLogX	for migration/response plots
MigrationLogY	for migration/response plots
MigrationTitleOffsetX	for migration/response plots
MigrationTitleOffsetY	for migration/response plots
MigrationZmin	for migration plots
MigrationZmax	for migration plots
ResponseZmax	for response plots
ResponseZmin	for response plots
PlotSystematicMigrations	if set to TRUE will plot migration/response plots for all systematics
MigrationText	if set to TRUE will show numbers in migration/response matrix plots
DivideByBinWidth	if set to TRUE will divide the bin content by bin width in the final plot with unfolded data
DivideByLumi	if set to a value > 0 will divide the bin content by the provided value (default: set to -1, disabled)
UnfoldNormXSec	if set to TRUE the result will be in the form of normalized cross-section, or relative cross-section, $\sigma(\text{bin}-1)/\sigma(\text{tot})$: the total cross-section is fitted as nuisance

Option	Function
	parameter, and the relative cross-section in one of the bins (selectable with the option <code>UnfoldNormXSecBinN</code>) is obtained as 1 - the other relative cross-sections (error is propagated)
<code>UnfoldNormXSecBinN</code>	in case <code>UnfoldNormXSec</code> is set to TRUE, one can specify which bin to obtain as a function of the cross-sections in the other bins (if not set, the last bin of the truth distribution will be selected)
<code>AlternativeAsimovTruthSample</code>	Can be used to create Asimov dataset by folding alternative (non-nominal) truth sample that is provided to get the reco distribution for the signal.
Expressions	a way to correlate the unfolding norm factors with other norm factors (other unfolding ones or not); analogous to the NormFactor option Expression, but accepts a list of expressions, with this format <code><norm-factor-1>=<expression>:<dependencies>, <norm-factor-2>=<expression>:<dependencies></code> [example: <pre>"Unfolding_Bin_002_mu"="0.5*(Unfolding_Bin_001_mu+Unfolding_Bin_003_mu)": "Unfolding_Bin_001_mu[-100,100], Unfolding_Bin_003_mu[-100,100]" , "Unfolding_Bin_005_mu"="Unfolding_Bin_004_mu":Unfolding_Bin_004_mu[-100,100]]</pre> (NB: mandatory usage of quotation marks in case of expressions with more that one argument, as in the example). If you want to use some parameters of the truth histogram, e.g. the truth bin yield of bin 2, you can add the following string to the expression: <code>@Yield2</code> - this will be replaced on the fly with the value from the histogram. The supported replacement strings are: <code>@YieldX</code> (BinContent), <code>@WidthX</code> (BinWidth), <code>@EdgeX</code> (BinLowEdge), <code>@CenterX</code> (GetBinCenter) and <code>@Integral</code> (Integral of the histogram). The <code>x</code> value has to be an integer index of the bin, starting from 1. Example: Expressions: <pre>"Unfolding_Bin_001_mu"="@Yield1*Unfolding_Bin_002_mu:Unfolding_Bin_002_mu[1, -10, 10]</pre>
<code>UnfoldingChi2Type</code>	can be set to <code>EMPTY</code> (default), will not show χ^2 values for unfolded data and MC in the unfolding plot. If set to <code>CHI2</code> , will

Option	Function
	show the χ^2 values, <code>CHI2/NDF</code> will show χ^2 /NDF values and <code>PROB</code> will show the probability. In all cases, all three values are printed in the terminal
UncertaintyBreakdownRange	Manually set the range of the uncertainty breakdown plot (uses -range, range). If not set, an automatic range is used.

TruthSample block

Option	Function
Title	for plots
LineColor	for plots
LineColorRGB	for plots
LineStyle	for plots
LineWidth	for plots
TruthDistributionPath	folder path for truth distributions
TruthDistributionFile	file path for truth distributions
TruthDistributionName	name of the histogram in the file for truth distributions
UseForPlotting	Can be set to <code>TRUE</code> (default) of <code>FALSE</code> , tells the code if the given truth distribution will appear on the final plot with unfolded data
UnfoldingName	String representing the Unfolding block. This is needed in case of multiple Unfolding blocks to associate the TruthSample to the Unfolding block